nature ecology & evolution

Article

https://doi.org/10.1038/s41559-024-02491-9

Morphological innovation did not drive diversification in Mesozoic–Cenozoic brachiopods

Received: 16 February 2024

Accepted: 28 June 2024

Published online: 25 July 2024

Check for updates

Zhen Guo [1], Michael J. Benton [2], Thomas L. Stubbs [3] & Zhong-Qiang Chen [1]

Over long spans of geological time, various groups of organisms may wax and wane, experiencing times of apparent success and contraction. These rises and falls are often said to reflect either opportunities created by climate change or the relative success of innovative characteristics. Phylum Brachiopoda was one of the most successful marine clades before the Permian/Triassic mass extinction (PTME), but after this event, they became marginal components of marine communities through to the present day. How brachiopod morphological innovations reacted to swiftly declining diversity has long remained poorly understood. Here we analyse morphological evolution over the 300 Myr (Permian–Quaternary) history of the four major Mesozoic–Cenozoic brachiopod orders (Terebratulida, Rhynchonellida, Spiriferinida, Athyridida). Unexpectedly, their disparities reached or exceeded pre-PTME levels, but were decoupled from generic richness, which was generally low. Distribution of taxa in morphospace and shifts in centroid indicate that all four orders exploited new morphospaces when adapting to post-Permian environments. A comparison of morphospace occupation and diversity evolution suggests that the high extinction rate of brachiopods and the limited diversification of new forms may have accounted for the depauperate nature of modern-day brachiopods.

Biologists and palaeobiologists often refer to 'success' in evolution, by which they mean the achievement by a clade of high species richness, high abundance and great breadth of morphological–functional adaptation. Among modern organisms, mammals, birds, molluscs and angiosperms might be said to be successful based on these metrics. However, turtles, crocodilians, brachiopods and ginkgos might be said to be less successful because of currently lower species numbers, lower global abundance and more restricted functional and ecological diversities. In comparing 'successful' and 'unsuccessful' clades, comparative studies often select pairs of close relatives, such as crocodilians and birds, that share a single common ancestor, in this case, some 252 million years ago (Ma) in the Triassic, and ask why birds today comprise more than 10,000 species yet their sister clade, crocodilians, include ~25 extant species[1–3]. Among marine organisms, brachiopods and bivalves are often taken as a pairing[4–10], not because they shared a close common ancestor but because of many shared anatomical and functional features[4]. The classic story has been that brachiopods dominated seabeds until the Permian/Triassic mass extinction (PTME), ~252 Ma, and then bivalves took over in the Triassic[11]. This has frequently been explained by the substantial loss of ecospace by brachiopods to bivalves[4,6,7,9], and the current marginal role of brachiopods in world oceans seems to bear this out. But is this true?

[1]State Key Laboratory of Biogeology and Environmental Geology, China University of Geosciences (Wuhan), Wuhan, China. [2]School of Earth Sciences, University of Bristol, Bristol, UK. [3]School of Life, Health and Chemical Sciences, The Open University, Milton Keynes, UK. e-mail: zhong.qiang.chen@cug.edu.cn

https://doi.org/10.1038/s41559-024-02491-9

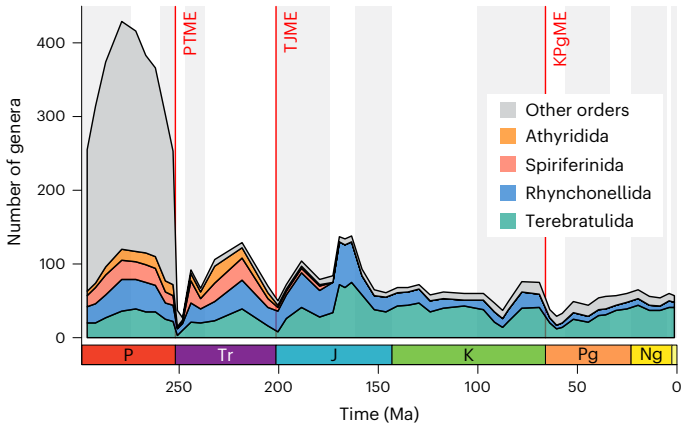


Fig. 1 | Diversity of Permian–Quaternary brachiopods. The four orders analysed in this study comprise the majority of the post-Permian brachiopod diversity. P, Permian; Tr, Triassic; J, Jurassic; K, Cretaceous; Pg, Palaeogene; Ng, Neogene, PTME, Permian/Triassic mass extinction; TJME, Triassic/Jurassic mass extinction; KPgME, Cretaceous/Paleogene mass extinction.

Phylum Brachiopoda originated in the Cambrian, survived the 'Big Five' mass extinctions[12,13] and persists in today's oceans[14]. They were the most abundant benthic invertebrates occupying most habitats in oceans before the Mesozoic. After the PTME, brachiopods lost diversity dominance and became marginal components of marine communities in Mesozoic–Cenozoic (Meso–Cenozoic) oceans[15–19] (Fig. 1). The remarkable switch from brachiopods to bivalves across the PTME and the permanent reduction in diversity of Meso–Cenozoic brachiopods have long intrigued palaeobiologists and biologists[4–7,9,10,20–23]. However, most previous studies interpreted this switch in terms of taxonomic diversity alone and assumed that brachiopods lost out in some way in competition with bivalves. A consequence of such a view would be that post-Permian functional diversity of brachiopods, represented by morphospace occupation, should be much reduced or show little evidence of innovation and novel adaptation, suggesting that brachiopods were squeezed into a reduced number of perhaps marginal niches by the success of the bivalves in and after the Triassic.

Morphological disparity, which describes the diversity of morphology in a clade (that is, 'morphological diversity'), is an important complement to taxonomic diversity to depict the evolutionary dynamics of organisms[24]. Although the two metrics may be coupled or decoupled through time[25–32], it is often seen that morphological innovation brings functional innovation and ecological opportunities to animals, paving the way for the upcoming taxonomic diversification[33–36]. In the history of the Earth, many clades rapidly accumulated great diversity at the beginning of their evolutionary history or swiftly recovered after mass extinctions, along with the expansion of their morphological range (that is, the early burst model)[27,37]. However, there are also some clades whose diversity never returned to pre-extinction levels such as the post-Permian brachiopods, and whether their apparent lack of success was limited by their morphological disparity remains unknown.

Although discussed for a long time, the evolution of brachiopod morphology is poorly understood[38]. Previous studies on brachiopod disparity generally focused on one of the relatively small articulate[39–41] or inarticulate orders[42,43]. A previous study[38] investigated the Phanerozoic disparity of articulate brachiopods and unravelled a partially recovered and volatile disparity among post-Permian taxa, but the numbers of taxa sampled and the characters analysed were rather small.

Here we compile a large, comprehensive morphological dataset and analyse disparity over the ~300 Myr (Permian–Quaternary) history of >1,000 genera belonging to four major orders of Meso–Cenozoic brachiopods (Terebratulida, Rhynchonellida, Spiriferinida and Athyridida) (Extended Data Fig. 1). Their total genus richness accounts for more than 95% of all post-Permian brachiopods, thus providing a nearly complete picture of morphological evolution of the entire phylum. This study aims to provide new insights into several questions. First, we used two disparity metrics, sum of variances (SOV) and sum of ranges (SOR), to quantitatively measure overall morphological variations of the Meso–Cenozoic brachiopods and to assess their coupling or decoupling relationships with diversity measures represented by genus richness. Second, we examine how the PTME, Triassic/Jurassic mass extinction (TJME) and Cretaceous/Palaeogene mass extinction (KPgME) and their recovery processes regulated morphological evolution of the post-Permian brachiopods. Finally, covariation of disparity, biodiversity and morphospace occupation offers new insights into the drivers responsible for the depauperate nature of present-day brachiopods.

Results

Disparity and diversity through time

The Terebratulida, characterized by a loop in the shell, is the largest order (in terms of taxonomic diversity) among post-Permian brachiopods and is also the most diverse brachiopod clade in today's oceans[44]. Our analyses reveal that terebratulide disparity (SOV) shows an overall increasing trend from the Permian to the Quaternary and that this is decoupled from the overall diversity of the order, which has peaks and troughs, without a clear upward trajectory.

Terebratulide disparity and diversity fluctuated through time (Fig. 2a). Disparity (SOV) shows major drops occurring in the Early Triassic, Middle Jurassic, early Late Cretaceous and Palaeocene. After each decline and in the Middle–Late Triassic, Late Jurassic, late Late Cretaceous and Eocene, the SOV rebounded rapidly to the pre-drop levels or even higher. The SOV is rather low in the Middle Jurassic; however, this low value is not caused by the small morphological range, but rather by the rapid diversification of the superfamily Loboidothyridoidea[44]. Although the members of this superfamily possess diverse external characters, their internal morphologies show little variance (from a macroscopic perspective of the entire order), and these forms therefore gather in a small region in the upper-left corner of the morphospace, pointing to a low SOV value (Fig. 3a). Similarly, terebratulide SOR also exhibits a generally increasing trend (Extended Data Fig. 2) although the trend is more subtle. The broad confidence intervals (due to the small sample size after rarefaction) make the significance difficult to confirm. After the PTME and TJME, terebratulide diversity surged to a very high level in the Middle Jurassic and then returned to the Permian level during the Late Cretaceous. The Cenozoic witnessed a steady increase in diversity, but present-day diversity never returned to the mid-Jurassic level.

The crura-bearing Rhynchonellida is the second-largest order among the Meso–Cenozoic brachiopods. Their disparity (SOV) showed an overall increasing trend from the Triassic to the Jurassic and then declined in the Early Cretaceous. From the Late Cretaceous to the Quaternary, a clear trend was not exhibited, and the confidence intervals were very wide (Fig. 2b). The evolution of the SOV contrasts markedly with diversity, which massively decreased following the Jurassic and remained constantly low to the present day, where this order is just a small subset of living brachiopods. More specifically, the rhynchonellide SOV peaked in the Early–Middle Permian, Late Triassic, Late Jurassic–earliest Cretaceous and Late Cretaceous, and dropped in the Late Permian, Middle Jurassic, early–middle Cretaceous and Palaeocene. The SOR of this order shows a similar trend to the SOV (Extended Data Fig. 2), except for even lower values from the mid-Cretaceous to the present day.

The Spiriferinida, an order comprising spire-bearing and punctate taxa, includes far fewer genera than the first two orders, yet it is one of the major components of Triassic brachiopod faunas. Their taxonomic diversity, SOV and SOR were all severely affected by the

https://doi.org/10.1038/s41559-024-02491-9

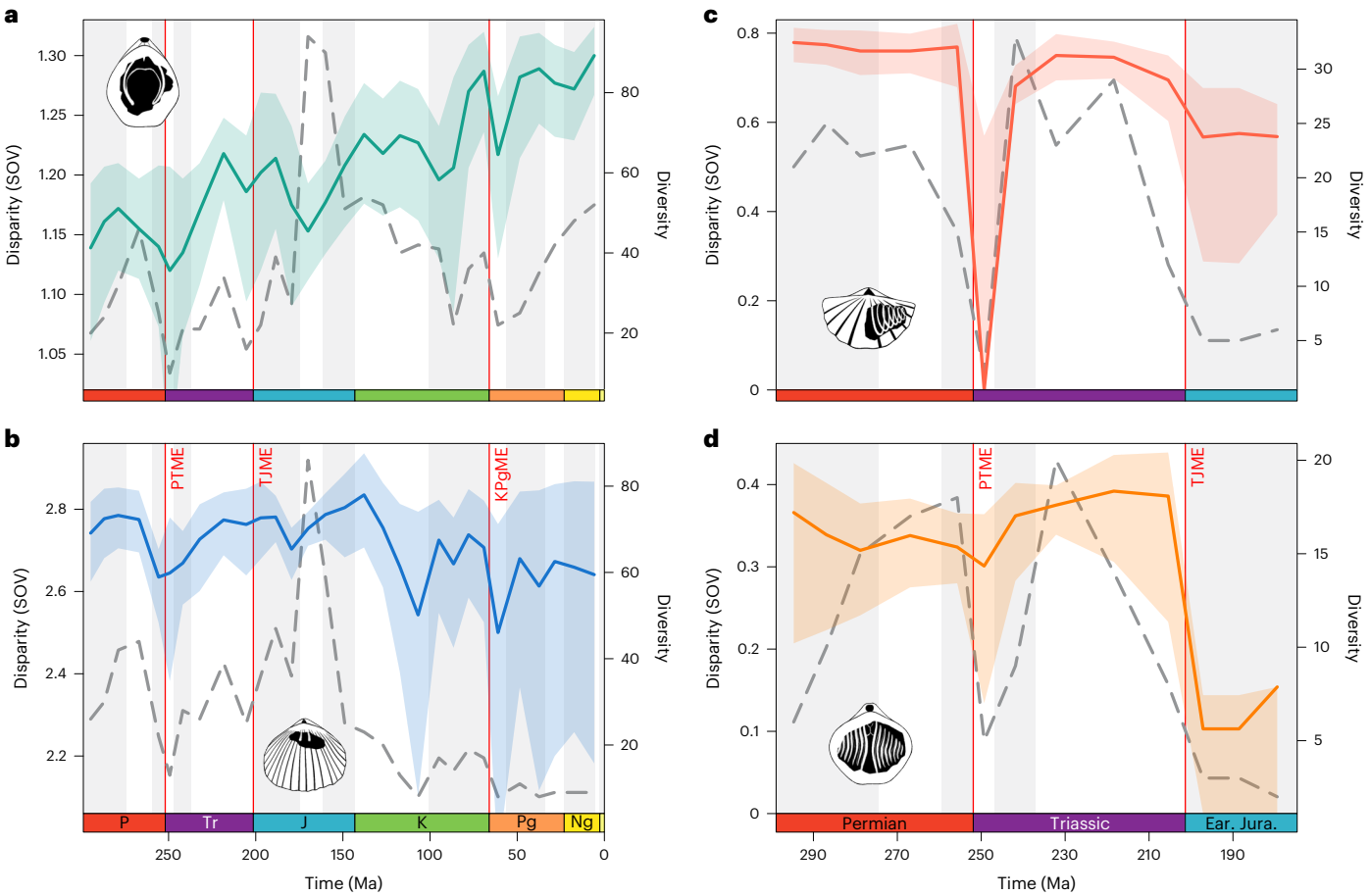


Fig. 2 | Disparity and diversity of four brachiopod orders through time. Disparity is measured by the SOV. The solid lines represent the median value of the bootstrapped disparity. The envelopes indicate 95% confidence intervals (0.025 and 0.975 quantiles of bootstrapped values). Diversity (number of genera) is represented by the dashed lines. **a**, Terebratulida. **b**, Rhynchonellida. **c**, Spiriferinida. **d**, Athyridida. Ear. Jura., Early Jurassic.

PTME (Fig. 2c and Extended Data Fig. 2), but rapidly rebounded and finally reached the pre-extinction value in the Middle to Late Triassic. Diversity declined rapidly in the Rhaetian, before a further hit by the TJME. Disparity also followed this trend and gradually declined to a lower level. In Early Jurassic, these three metrics did not recover until this order was wiped out in the Toarcian[45].

The Athyridida is another spire-bearing group that became extinct at the Toarcian Oceanic Anoxic Event (Early Jurassic) along with Spiriferinida[45,46]. Athyridide SOV was almost unchanged through the Permian/Triassic transition despite a notable drop in diversity. Their SOV rose through the Triassic, sharply declined after the TJME and then remained at low levels towards the final demise of this order in the Early Jurassic (Fig. 2d). The SOR shared a similar pattern to the SOV, that is, a high value in the Permian and the Triassic and an extremely low value in the Jurassic (Extended Data Fig. 2). After the PTME, athyridide diversity did not recover until the Carnian of the Late Triassic when it surged prominently, followed by major drops in the Rhaetian (late Late Triassic) and TJME, and then rather low levels in the Jurassic (Fig. 2d).

Morphospace occupation through time

The shifts of morphospace occupation in brachiopod genera can be directly observed in the morphospace (Fig. 3 and Extended Data Figs. 3–6). However, for some datasets, the first few morphospace axes explain only a low proportion of variance and a great amount of information is hidden on the other axes (Extended Data Fig. 7). Therefore, we also constructed a 'morphospace centroid space' to complement our visualization of the results (see Methods for an extended description; Fig. 4 and Extended Data Fig. 8), in which the distribution of time bins in the centroid space shows the similarity of morphological composition in different bins. Overall, temporal trends of morphospace occupation show continued innovation through time despite reductions in diversity at major extinction events.

Terebratulides are unevenly distributed in morphospace (Fig. 3a). The morphospace areas occupied by short-looped and inner-hinge-plates-lacking taxa and by long-looped taxa have higher density, while other areas are less crowded (Fig. 3a). From the Triassic to the Jurassic, taxa in these two areas diversified greatly, whereas those in the upper-central area gradually disappeared. Taxonomically, this pattern reflects the diversification of two major descendants of the Dielasmatoidea (that is, the short-looped Loboidothyridoidea and long-looped superfamilies such as Zeillerioidea and Laqueoidea) and the extinction of the Dielasmatoidea itself[44]. The Jurassic to Cenozoic occupation trend is characterized by the invasion of morphospace areas occurring in the lower part of the plot, which is occupied by taxa having a septal pillar and lacking hinge plates. The centroid space of Terebratulida shows that the centroids are widely separated in the Permian, Triassic and Jurassic–Cenozoic intervals, indicating different morphological compositions (Fig. 4a). The Jurassic and Cenozoic centroids are also separated, but they are bridged by the Cretaceous ones, suggesting that the morphology of terebratulides has gradually changed to the current form since the Jurassic. The greatest shift in centroids of morphospace occupation occurred in the Early Triassic (Fig. 4b). Although the sample size is low in the Early Triassic, this shift is significant according to the result of the permutational multivariate

https://doi.org/10.1038/s41559-024-02491-9

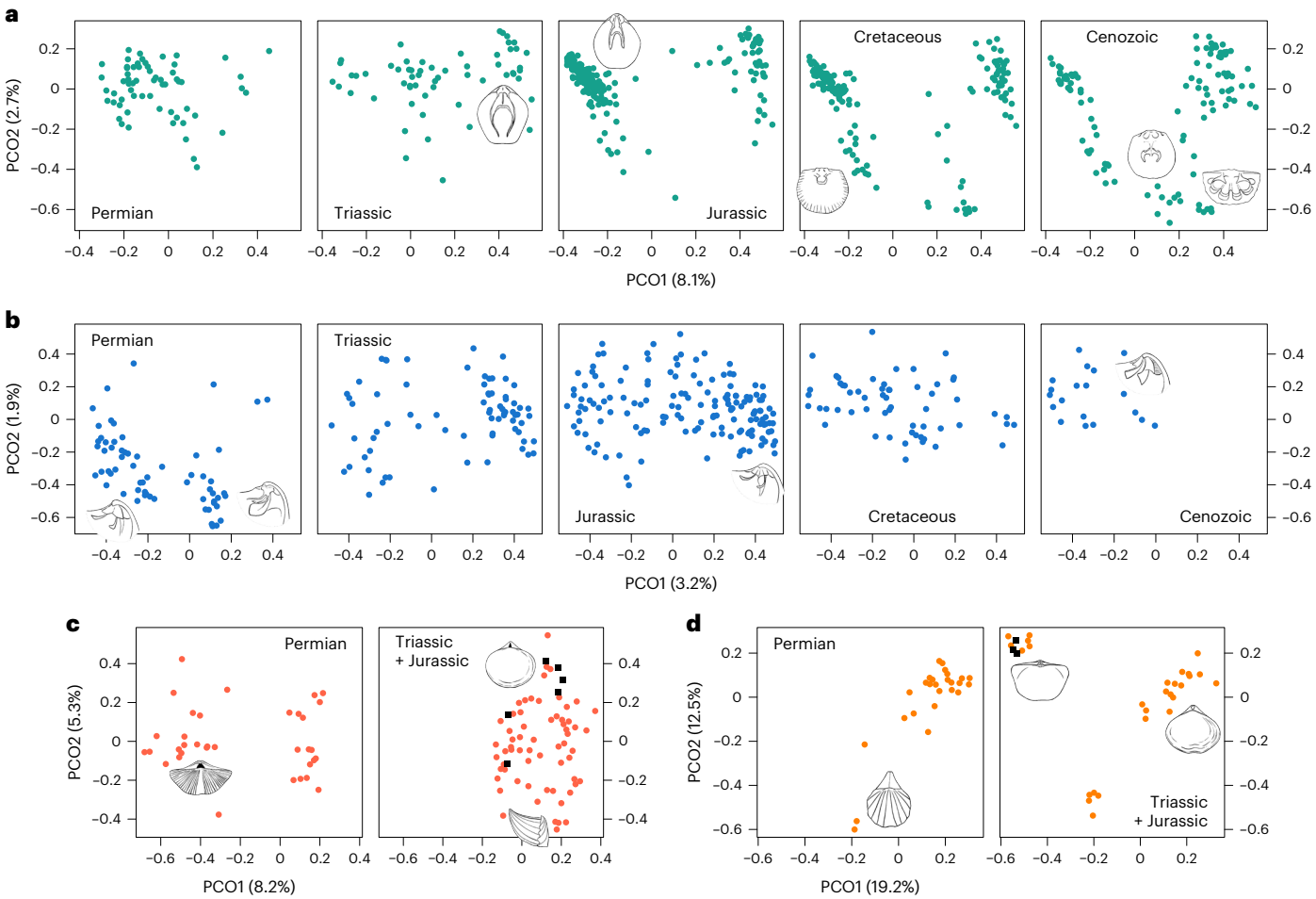


Fig. 3 | Morphospace occupation of the four brachiopod orders through time. **a**, Terebratulida. **b**, Rhynchonellida. **c**, Spiriferinida. **d**, Athyridida. Morphospaces are constructed by the first two principal coordinates (PCO1 and PCO2). The number in the axis labels indicates the percentage of variances explained by the axis. The black cubes in **c** and **d** represent Jurassic taxa. The illustrations (from left to right) in **a** represent long-looped, deltiform-looped and inner-hinge-plates-lacking, ring-like-looped, septal-pillar-bearing and loop-incomplete, and septal-pillar-bearing and loop-floor-fused terebratulides, respectively. Those in **b** represent hinge-plates-fused, camarophorium-bearing, septalium-bearing-raducal, and septifal rhynchonellides, respectively. Those in **c** represent spiriferiform and median-septum-lacking, reticulariform, and cyrtiniform spiriferinides, respectively. Those in **d** represent multicostate and punctate, concavo-convex, and bivoncex athyrididdes, respectively. For morphospace occupation variation in finer time bins, see Extended Data Figs. 3–6.

ANOVA test ($P < 0.05$) (Supplementary Information). Moreover, obvious shifts in centroid also occurred in the aftermaths of TJME and KPgME (Fig. 4b), indicating that mass extinctions may have altered morphological compositions of terebratulides.

For rhynchonellides, variations of morphospace occupation indicate that the previous areas were lost when new areas were explored. For instance, the Triassic–Jurassic septalium-bearing taxa fully exploited the middle-right area and the Jurassic–Cenozoic septifal taxa developed more in the left half region in the morphospace (Fig. 3b). Meanwhile, the lower part of the space (camarophorium-bearing taxa and hinge-plates-fused taxa) was unoccupied after the Permian, and the right areas were gradually lost from the Cretaceous to the Cenozoic. The centroid space of this order shows that the Permian centroids are clearly distinguished from those of other periods (Fig. 4c). From the Triassic to the Cenozoic, the centroid moved along the first principal coordinate axis, indicating a gradual transformation from Triassic morphologies to present-day ones. The most prominent shift in the centroid occurs in the Early Triassic despite rather high values also occurring in the Palaeocene and the Eocene, which, however, could not be reliable owing to the limited sample size (Fig. 4d and Supplementary Information).

The Spiriferinida broadened the morphospace and occupied new areas in the Triassic[41], while the previous morphospace occupied by the Permian superfamily Syringothyridoidea disappeared forever (Fig. 3c). By contrast, the Early–Middle Triassic athyrididdes re-thrived in the areas previously occupied by the Permian elements of the same order, and then greatly expanded to the upper-left corner by evolving concavo-convex forms in the Koninckinidina (Fig. 3d). Both spiriferinides and athyrididdes did not invade new morphospace in the Jurassic, and the Jurassic morphospaces are only a subset of their Triassic ones (Fig. 3c,d). Their morphospace centroids are separated from each other in the Permian, Triassic and Jurassic (Fig. 4e,g). The Spiriferinida shows a distinct centroid shift after the PTME (the Early Triassic shift is doubtful owing to the small sample size) (Fig. 4f), while the Athyridida shows a larger centroid shift after the TJME (Fig. 4h).

Discussion

Effect of mass extinctions on brachiopod morphology

Of the three Mesozoic mass extinctions, the PTME not only depleted diversity but also regulated the distribution of morphospace (Figs. 1, 3 and 4), and thus reshaped brachiopod macroevolution[14,17,18]. The Palaeozoic orders (Productida, Spiriferida, Orthotetida, Orthida and Dictyonellida) were not analysed in this study, but it is obvious that these high-level groups and the associated morphologies were permanently

https://doi.org/10.1038/s41559-024-02491-9

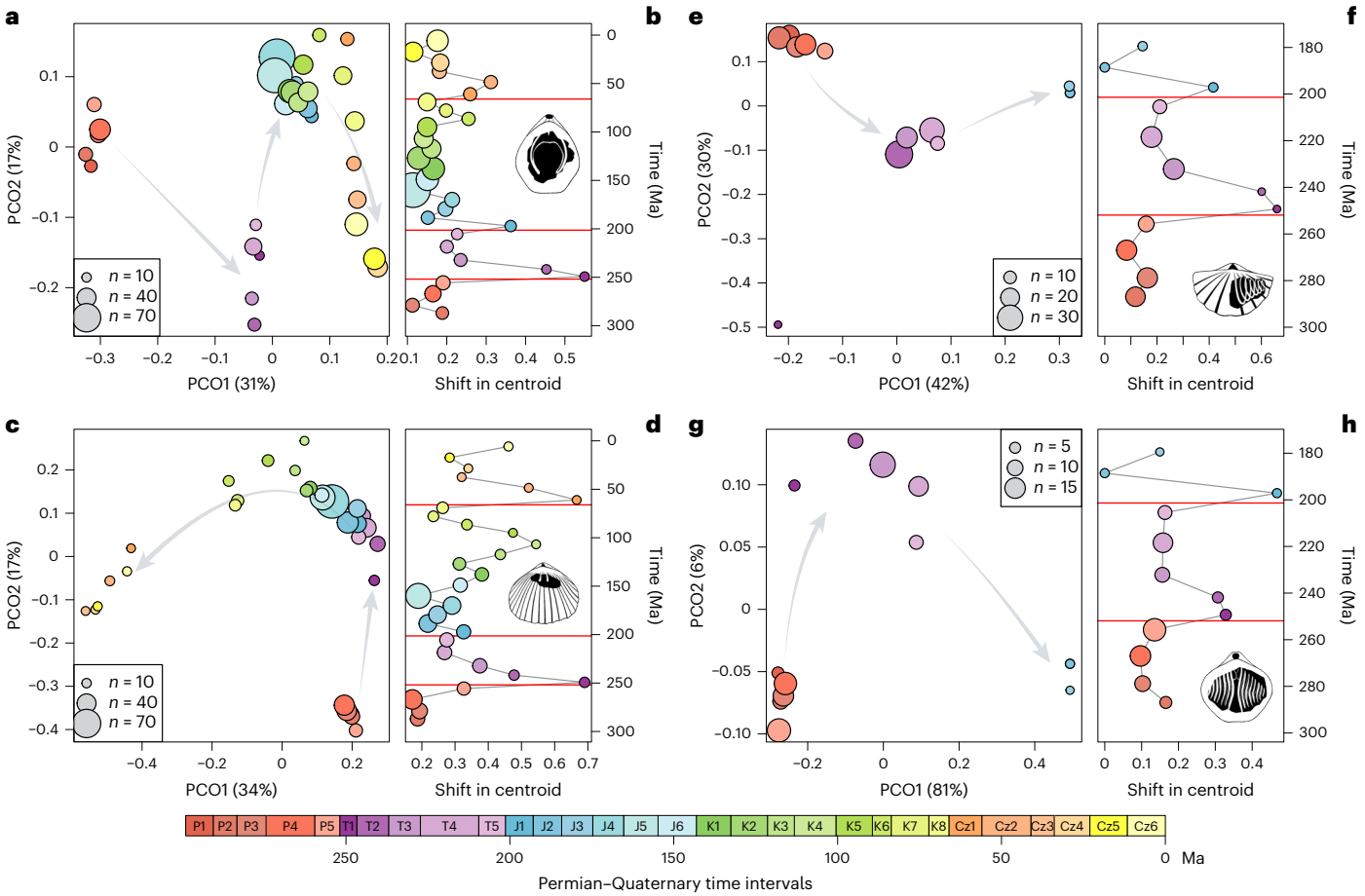


Fig. 4 | Morphospace centroid spaces and shifts in centroid position of four brachiopod orders. a,**b**, Terebratulida. **c**,**d**, Rhynchonellida. **e**,**f**, Spiriferinida. **g**,**h**, Athyridida. **a**,**c**,**e**,**g**, Centroid spaces are visualizations (using PCOA) of the distance matrix of centroids in morphospace and are constructed from the first two principal coordinates (PCO1 and PCO2). Numbers in the axis labels indicate the percentage of variance explained. The grey arrows indicate the trend of centroid movement in the centroid space. **b**,**d**,**f**,**h**, Shifts in centroid indicate the distance between centroids of one bin and its previous bin in the morphospace. In the plots of centroid space, the size of a dot is related to the diversity in that time bin. In the plots of centroid shifts, the size of a dot is proportional to the lower value of diversity in that bin and diversity in its previous bin (for example, the size is proportional to 20 if diversities in that bin and its previous bin are 30 and 20, respectively). The larger the dot, the more reliable the shift because of the larger sample size. The red lines represent mass extinctions. For definitions of the time bin names, see Methods. See Extended Data Fig. 8 for centroid spaces with bin names labelled.

eliminated by the PTME. It is noteworthy that although morphological compositions of the four surviving orders were altered by this mass extinction, their disparity values showed less long-term effect from the PTME. The disparity and diversity of both the Spiriferinida and Athyridida recovered to pre-extinction levels in the Triassic, and morphological innovations in Rhynchonellida and Terebratulida emerged and their disparities returned to pre-extinction levels in the early Mesozoic. By contrast, the TJME had little effect on the Rhynchonellida and Terebratulida, which both proliferated after the crisis. The other two orders (Spiriferinida and Athyridida) suffered severe impacts from the Late Triassic events and TJME and fell into the state of 'dead clade walking'[47,48] in both diversity and disparity, before their disappearance in the Early Jurassic. The TJME therefore had more prolonged impact on disparities of Spiriferinida and Athyridida than the PTME. The KPgME did not substantially reshape the morphological evolution of terebratulides and rhynchonellides. It only accelerated the transition of morphological composition from the Cretaceous ones to the modern ones (Figs. 3 and 4).

In addition to these mass extinctions, other environmental events (for example, oceanic anoxia events in the Jurassic and the Cretaceous) also caused diversity perturbations of brachiopods[49,50]. However, they did not show notable effects on the morphological evolution of the four orders studied (Figs. 3 and 4). A more detailed study in finer time bins might reveal their influences.

Morphological innovation in post-Permian brachiopods

Variations of the SOV, SOR and morphospace occupation show that these four orders exhibit four different morphological evolution patterns. The Terebratulida has steadily broadened new morphospace areas since the Triassic, and only a small part of the preoccupied areas was lost. By contrast, the Rhynchonellida explored new morphospace in the Triassic and Jurassic, and frequently retreated from some previously occupied areas. The Spiriferinida rapidly diversified in morphology in the Triassic and took over a great part of the morphospace of the Palaeozoic order Spiriferida[41], while the Triassic athyridides largely inherited their Permian precursors' morphologies except for a major step of innovation introduced by the new suborder Koninckinidina (Fig. 3).

Disparity analyses of many animal clades show pervasive early high disparity; that is, animal clades tend to reach their maximum disparity relatively early in their evolutionary histories[27,37]. The pre-Permian taxa of the four orders were not analysed in this study, but according to the number of higher-level groups (that is, suborders, superfamilies)[51], such an early burst was not observed in at least three of these four Meso–Cenozoic orders. The Terebratulida and Spiriferinida reached

https://doi.org/10.1038/s41559-024-02491-9

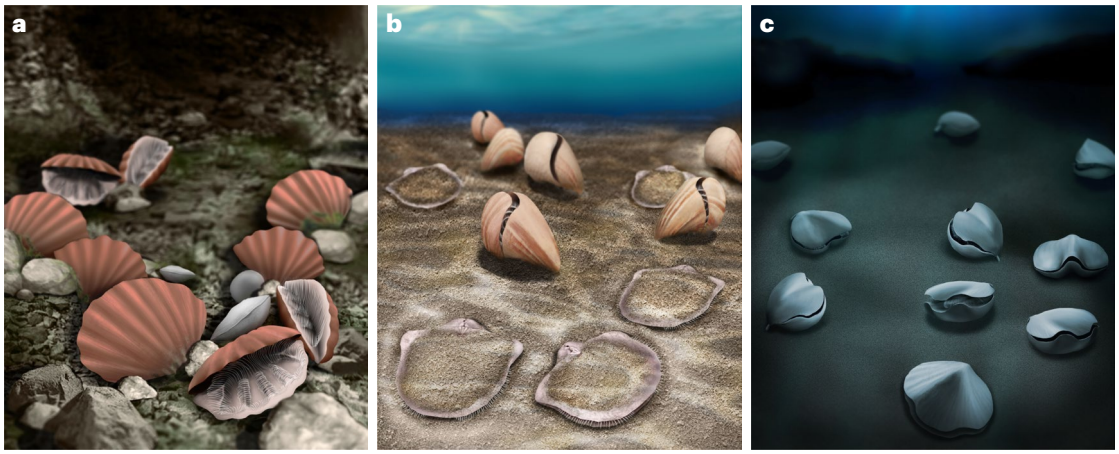


Fig. 5 | Reconstructions of some post-Permian brachiopods, showing adaptations to certain environments. a, Small-sized terebratulides and tiny rhynchonellides living in shallow-water caves. **b**, Concavo-convex athyridides and spiriferinides with a high interarea adapted to soft substrates. **c**, Smooth and weakly ornamented rhynchonellides and terebratulides living in deep environments. Figure drawn by Shunyi Shi.

their highest disparities in the Cenozoic and Late Triassic, respectively. The Athyridida, typical of Palaeozoic orders, had high diversity in the Palaeozoic, but the survival of two suborders through the PTME and the emergence of a new suborder in the Triassic possibly elevated its disparity to a higher level. Rhynchonellides, originating in the Ordovician, were much more diverse in the Devonian than in the Mesozoic and evolved into several superfamilies at that time. Because we did not study the Devonian taxa, a direct comparison is not yet possible. In terms of the entire phylum Brachiopoda, the maximum disparity possibly occurred in the late Palaeozoic[38], about midway through its evolutionary history. As a result, early high disparity was possibly not very pervasive in brachiopods.

Each of the four orders showed its own pattern of morphological evolution, but they all explored new morphospace areas. Rigorous statistical analysis of morphology and associated environments is difficult owing to the lack of data for many genera, but there are several examples showing adaptations of post-Permian brachiopods to new environments (Fig. 5). For instance, the Athyridida had the most conspicuous innovation in the emergence of the Koninckinidina, members of which are readily distinguished by having unique concavo-convex shells that are similar to the Palaeozoic strophomenatans, indicating an infaunal lifestyle adapted to soft substrates[52] (Fig. 5b). The Spiriferinida possessed highly varied shell outlines and delthyrium covers during the Triassic, indicating different ways to stabilize the shell bodies[41] (Fig. 5b). The rhynchonellides also exhibit co-evolution between morphology and environments[50,53]: the ribbed and raducal group often inhabits epicontinental seas, while the smooth and weakly ornamented taxa with arcuiform and septifal crura prefer to dwell in epioceanic or deeper environments[50]. Together with smooth rhynchonellides, terebratulides inhabiting deeper habitats also tend to be smooth-shelled and possess a unisulcate anterior commissure[54,55] (Fig. 5c). Another important morphological change of the Terebratulida is the retention of a septal pillar (Fig. 3a), like those in the Platidioidea, Kraussinoidea and Megathyridoidea. In present-day oceans, many megathyridoids live in cryptic habitats, such as shallow-water caves or undersides of coral colonies[56] (Fig. 5a), and it was assumed that the low metabolic rates and low-energy lifestyles help these micromorphic brachiopods to survive in such nutrition-poor environments[57].

All lines of evidence indicate that after undergoing an evolutionary bottleneck at the PTME, post-Permian brachiopods struggled to expand into new morphospaces, especially the vacated ecospaces of the early Triassic, like the behaviours of molluscs and other elements of the 'modern evolutionary fauna'[58–60].

It was hypothesized that brachiopods possess relatively less flexibility to modify their basic body plans than bivalves[5], probably limiting the ecological success of brachiopods. Unfortunately, a direct comparison between these two clades is not yet possible owing to the unavailability of the long-term disparity evolution trend of bivalves. However, the rise of a new brachiopod order, the Thecideida (not included in our analysis owing to an incomplete fossil record) in the Late Triassic, possibly does not support this hypothesis. Members of Thecideida have a small size, a well-developed ventral interarea with a pseudodeltidium, a lid-like dorsal valve and a ventral cicatrix allowing the cemented lifestyle. Internally, thecideides possess a prominent cardinal process, a pair of well-developed inner socket ridges, a variously developed median septum and pair(s) of brachial lobes. These morphological features are so unique that only few characters are shared with other orders, making their phylogenetic relationships with other brachiopods controversial[61,62]. Also debated is the systematic position of the suborder Koninckinidina, although they are currently assigned to the Athyridida[63]. These higher-level innovations provide further evidence for the evolvability in the post-Permian brachiopods.

Decoupling between morphospace and diversity evolutions

Both morphospace occupation and diversity evolution trends of brachiopods clearly show a decoupling of the two. Apparently, morphospace increase often did not drive prominent diversification of Meso–Cenozoic brachiopods (Figs. 2 and 3). We propose that two kinds of mechanisms that are related to the extinction and the origination process, respectively, account for such decoupled evolution between morphospace and diversity. Firstly, as essential components of the 'Palaeozoic evolutionary fauna'[64], brachiopods intrinsically had a higher extinction rate than the 'modern evolutionary fauna' such as bivalves[9,10,65]. Brachiopods were affected by various-scale extinction events more severely than molluscs, and many clades failed to survive to the present day. By contrast, most bivalve families are extant, meaning that their statistical risk of extinction is less than that of brachiopods through time[66]. For instance, the Spiriferinida and Athyridida successfully recovered in the Triassic, but they experienced a high extinction rate in the Late Triassic and the TJME[45], and finally became extinct in a relatively small extinction event, the Toarcian Oceanic Anoxic Event. The Jurassic radiation of long-flanged deltiform terebratulides (superfamily Loboidothyridoidea) and raducal rhynchonellides (superfamilies Rhynchonelloidea and Hemithiridoidea) are good examples of adaptive radiation[67]. These forms were abundant and widely distributed in various settings in the Middle Jurassic[53], making this the highest level after the PTME for diversity of the entire phylum (Figs. 1 and 3).

However, the rather high extinction rate in the Middle–Late Jurassic[10] resulted in a drop in their diversity. The vast majority of families did not survive into the Cenozoic and thus contributed little to the subsequent diversification of brachiopods. Why brachiopods are prone to extinction has been debated for a long time. Some researchers suggested that the fixed lophophore of both athyridides and spiriferinides is less advantageous during environmental stresses[45,68]. However, some spire bearers (for example, Athyridida, Spiriferida) were very successful in the late Palaeozoic after a great number of extinctions. Moreover, the rhynchonellides and terebratulides without a fixed lophophore also experienced a high extinction rate in the Middle–Late Jurassic. Other hypotheses have been proposed, such as the diversification of predators and competition with bivalves, but they cannot fully explain the observed patterns[10].

Limited origination of some new brachiopod forms is probably another cause for failure in diversification of post-Permian brachiopods. For instance, three new terebratulide superfamilies having a septal pillar in adults (Platidioidea, Kraussinoidea and Megathyridoidea) occurred in the Cretaceous and Cenozoic, evidently broadening morphospace occupation (Fig. 3a). Each superfamily, however, contains fewer than ten genera (including fossil and living forms)[44], indicating a rather low origination efficiency of these new high-level classification units. Moreover, the morphologically unique suborder Koninckinidina includes only nine genera[63]; the new order Thecideida, through an ~240 Myr evolution, has only about 40 genera[62]. The taxonomic numbers of these two new high-level classification units of Meso–Cenozoic brachiopods are much lower than their Palaeozoic morphological and ecological analogues (for example, Productida and Strophomenida). In other words, adaptive radiation was not observed in these brachiopod clades, reinforcing that morphological innovation is independent of ecological opportunities and evolutionary success[69,70].

Study limitations

In this paper, we analysed four major orders of brachiopods. Three other Meso–Cenozoic orders, including the articulate Thecideida and inarticulate Lingulida and Craniida, were not considered. Their ecological lifestyles are distinguished from the four analysed orders, but their fossil record is patchy and discontinuous, hampering a thorough study of morphological evolution. The Thecideida has highly varied internal structures[62], and the shell morphology of the Lingulida is much more complicated than previously thought[71–73]. These orders probably have different trends in morphological evolution from the four studied orders, thereby increasing the complexity of the evolution of the entire phylum.

Morphology closely relates to the ecology of brachiopods. However, as mentioned above, morphological disparity is not equivalent to ecological disparity and anatomical innovation does not always bring ecological breakthroughs[36,70]. Unlike other animal clades with multiple body parts, the shell of brachiopods is relatively simple. All structures must work coherently to ensure a normal life for a brachiopod. As a result, it is difficult to disentangle the function of a certain trait, and functions of many characters are speculative. If the connections among morphology, function and environment were better known, future independent analyses of ecological disparity and morphological disparity[36,74] might provide a deeper understanding of the waxing and waning of brachiopod diversity in Meso–Cenozoic oceans.

Concluding remarks

Although brachiopod diversity remained low and did not show great fluctuations in the Mesozoic and the Cenozoic, morphological evolution shows a far more dynamic pattern. The PTME significantly altered the morphological composition of brachiopods, but the disparity of surviving groups successfully rebounded or even exceeded the pre-extinction values. All orders exploited new areas in morphospace, and the Meso–Cenozoic morphology is not simply a subset of the Palaeozoic one. This evidence suggests that brachiopod evolvability was not prohibited by the 'modern evolutionary fauna'; instead, it was still rather active after the PTME. Nonetheless, not all new morphological forms succeeded in diversification; some new higher-level groups did not contribute much to the diversity of the phylum. Furthermore, the high extinction rate of brachiopods resulted in shorter longevity (stratigraphical duration) of brachiopod higher-level clades relative to bivalves, preventing the accumulation of diversity. In short, morphological innovation did not save the phylum from decline. Why brachiopods are prone to extinction and why some post-Permian forms did not have the same opportunity to diversify as their Palaeozoic morphological analogues are two questions that need further investigation.

Overall, the morphological evolution of the four orders showed very different patterns across the three mass extinctions in the studied interval: an extinction event may be morphologically selective (for example, TJME on athyridides) or nearly non-selective (for example, PTME on athyridides, KPgME on terebratulides) and the long-term impacts on disparity values may be weak (for example, PTME on the four orders) or strong (for example, TJME on athyridides and spiriferinides), showing the great heterogeneity among lineages and extinction events. This study highlights the importance of investigating morphological evolution as well as species (or generic) richness to understand the evolutionary history of a clade.

Methods

Fossil datasets

This study aims to investigate the morphological evolution and diversity decline of Meso–Cenozoic brachiopods; thus, the orders restricted to the Palaeozoic (for example, Productida, Orthotetida, Orthida, Spiriferida and Dictyonellida) were not considered. In the Mesozoic and Cenozoic, articulate brachiopods consist of five orders: Terebratulida, Rhynchonellida, Spiriferinida, Athyridida and Thecideida. Thecideida has a very rare fossil record and was not included in the compiled comparative dataset for this study. Inarticulate brachiopods (orders Lingulida and Craniida) are quite distinct ecologically and anatomically. However, they have a very low diversity after the Palaeozoic and their fossil records are discontinuous, preventing a complete and detailed analysis. Therefore, this study focused on the evolution of four orders: Terebratulida, Rhynchonellida, Spiriferinida and Athyridida, representing together >95% of total genus richness of all Meso–Cenozoic brachiopods; the study of these four groups is sufficient to understand the Meso–Cenozoic decline of brachiopods.

Fossil occurrences of Terebratulida, Rhynchonellida, Spiriferinida and Athyridida from the Permian to Recent (~298.89–0 Ma) were downloaded from the Paleobiology Database (PBDB) on 29 March 2024. All analyses herein were carried out at the genus level. The data cleaning and revision procedure followed ref. 10. Simply, (1) the name of each genus was examined, and only valid names were retained; (2) the temporal duration of each genus was checked, and doubtful records, namely those outside the stratigraphical range in well-curated datasets such as *Treatise on Invertebrate Paleontology*[51] and Sepkoski's compendium[16], were discarded; and (3) the ages of taxonomic occurrences were updated according to the *Geological Time Scale 2020*[75], using the fossilbrush R package[76]; occurrences with a high temporal uncertainty (>10 Myr) but not from an international stage were removed. In addition, the new Permian–Jurassic occurrences added in a previous study[10] were kept. Occurrences of 134 genera included in the *Treatise on Invertebrate Paleontology* but absent in the PBDB were also added to the final dataset compiled here.

Presence or absence tables of taxa from the Permian to the present day were made based on the fossil occurrence data. To ensure a roughly equal duration for each time bin, we divided the Permian–Quaternary time span into 30 intervals: P1, Asselian–Sakmarian; P2, Artinskian; P3, Kungurian; P4, Roadian–Capitanian; P5, Wuchiapingian–Changhsingian; T1, Induan–Olenekian; T2, Anisian–Ladinian; T3, Carnian; T4,

Norian; T5, Rhaetian; J1, Hettangian–Sinemurian; J2, Pliensbachian; J3, Toarcian; J4, Aalenian–Bathonian; J5, Callovian–Oxfordian; J6, Kimmeridgian–Tithonian; K1, Berriasian–Valanginian; K2, Hauterivian–Barremian; K3, Aptian; K4, Albian; K5, Cenomanian–Turonian; K6, Coniacian–Santonian; K7, Campanian; K8, Maastrichtian; Cz1, Palaeocene; Cz2, Ypresian–Lutetian; Cz3, Bartonian–Priabonian; Cz4, Rupelian–Chattian; Cz5, Aquitanian–Serravallian; and Cz6, Tortonian–Holocene. Except for some bins with very long durations such as Norian (T4), others have a span close to 10 Myr. The 'range-through' method was used to infill the presence or absence tables of genera, and diversity was calculated based on generic richness counts following their stratigraphical ranges. Thus, Carboniferous records were also used to extend the ranges of some Permian taxa. To eliminate the effect of 'the Pull of the Recent'[77], the present-day records of fossil taxa were not considered, and living genera without fossil records were also ignored.

Morphological datasets

The four orders are very different in both external and internal morphologies. Therefore, we analysed the morphological disparity of each order separately and devised specific character lists for each order (Supplementary Information). The characters of Spiriferinida were modified from those of a previous study[41]. Nevertheless, only external characters were included in the dataset of the previous study[41], so the internal structures of this order were added in the new dataset analysed. The characters of Rhynchonellida were selected and emended based on refs. 78,79. The characters of Terebratulida and Athyridida were inspired by ref. 80 and ref. 63, respectively.

In contrast to previous studies on brachiopod disparity[38], almost all major external (for example, outline, convexity, umbo features, ornamentation) and internal (for example, dental plates, hinge plates, loop-supporting structure) characters of shells that are commonly described in systematic studies were compiled in the dataset. Soft-body-related features were omitted because they are rarely known in fossil taxa. Cladistically non-informative characters (that is, autapomorphies, traits appearing in only one taxon) were included to comprehensively describe the morphological variation and retain unique innovations[81]. Characters are continuous (for example, shell width relative to shell length) or discrete, with the discrete characters being either ordered (for example, position of ventral foramen) or unordered (for example, type of anterior commissure). Some nested relationships are present in the characters (for example, the 'orientation of dental plates' is dependent on the 'presence or absence of dental plates'), so characters were subdivided into 'primary' and 'secondary' categories, which would have implications during the calculation of the distance matrices and stop secondary characters re-ranking the pairwise dissimilarities based on primary ones[82]. In total, 128, 81, 48 and 59 characters were coded for Terebratulida, Rhynchonellida, Spiriferinida and Athyridida, respectively. Detailed descriptions of characters are provided in Supplementary Information.

The character states of each genus were coded based on the adult morphology of its type species. If the type species was not perfectly known, another species under the same genus was selected. Ontogeny is vital in studying the evolution of brachiopods, and it is used as a key diagnosis in the higher-level classification units (for example, the complex loop development in Loboidothyridoidea versus the simple loop development in Terebratuloidea[44]). Nevertheless, ontogenetic variation is known in only a very small fraction of taxa[44]. Therefore, only the morphology of adults was considered in this study. Continuous characters were measured on one specimen of the type species, in most cases, the one shown in the *Treatise on Invertebrate Paleontology*. One may argue that such measurements are highly variable within one genus or one species (for example, the interarea height of *Liospiriferina* species). However, owing to the long time interval and wide taxonomic scope of our study, normally, greater differences appear between than within genera (for example, the low interarea of *Liospiriferina* compared with the high interarea of *Cisnerospira*).

Disparity analysis

The morphological datasets were used to calculate pairwise dissimilarity matrices using the maximum observable rescaled distance metric[83]. The alpha value applied to balance the contribution of primary characters and secondary characters was set to 0.5 as recommended[82]. Recent studies have shown that missing data can affect the position of taxa in morphospace[84,85]. Therefore, before the calculation of the distance matrices, taxa with more than 20% or 30% (20% for terebratulides and rhynchonellides; 30% for spiriferinides and athyridides) unknown characters (marked by a question mark) were discarded from the dataset. For terebratulides, taxa whose loop type is not revealed were also deleted because the loop structures have many morphological characters and these include the most important criteria for the classification of terebratulides. In the end, less than 10.5% of genera in these orders were discarded, and 529, 332, 104 and 54 genera were retained in the terebratulide, rhynchonellide, spiriferinide, and athyridide datasets, respectively (Supplementary Information). The diversity trajectories of the reduced datasets significantly correlate with that of the raw datasets (both raw and detrended diversities; Pearson's $r > 0.98$, $P < 0.05$; Extended Data Fig. 1). The dissimilarity matrices were arcsine square root transformed to make sure that the distances were normally distributed[83]. To construct morphospaces, principal coordinate analysis (PCOA) was applied to the transformed dissimilarity matrix using the pcoa function of the ape R package[86]. A Cailliez correction[87] was also used to deal with negative eigenvalues, at the expense of reducing the reported variance expressed by each axis[88].

There are multiple ways to evaluate the distribution of taxa in a morphospace and generate within-bin disparity time series, and each measurement has its advantages and disadvantages[89,90]. We used two popular metrics, the SOV and the SOR, to show the evolutionary trends of morphological disparity. The SOV describes the density of taxa within the occupied morphospace area, while the SOR reflects the total range of the area occupied. The SOR has some obvious drawbacks. It is sensitive to, and usually positively correlates with, sample size. Thus, rarefaction analysis is needed to compare the SOR of two samples[91], which may result in very wide confidence intervals if the sample size is low. Moreover, the SOR cannot reflect variation in the central part of the morphospace (for example, unoccupied central 'holes') and it is easily biased by taxa located in marginal areas of morphospace and outliers[89,90]. By contrast, the SOV is insensitive to sample size and outliers; therefore, it is widely used by many researchers although it may be biased by taxonomic over-splitting or clumping. In this paper, we prefer to use the SOV as an overall indicator of disparity. The SOR was calculated as a complementary index (Extended Data Fig. 2). Both the SOV and SOR were calculated from PCO scores of all axes. For each time bin, the median value and 95% confidence intervals were calculated based on 1,000 bootstrap replications[92]. For the SOR, because of its sensitivity to sample size, the number of taxa was rarefied to the lowest diversity of all time bins in the terebratulide and rhynchonellide analyses, and to the median diversity of all time bins in the spiriferinide and athyridide analyses. All disparity calculations were accomplished by the dispRity[93] and vegan[94] R packages.

Morphospace occupation

Differing from density (SOV) and size (SOR), the position of taxa in morphospace is another important aspect of morphological analysis[90]. The distribution of taxa in morphospace and variations in morphospace occupation are usually shown directly in the form of biplots of the major axes (Fig. 2 and Extended Data Figs. 3–6). However, when the first few PCO axes explain only a low percentage of the overall variation (such as the rhynchonellide morphospace in this study; Extended Data Fig. 7),

only a small proportion of total variation can be observed directly on the morphospace visualization and a great fraction of variation is hidden on other axes.

Here we used a centroid-related metric to evaluate the movement of morphospace occupation. The 'position of centroid' is an easy way to describe the position of taxa, which has been widely used by other researchers[31,74]. Normally, the distribution of taxa in the morphospace may be non-normal and complex, and therefore, some variation in morphospace cannot be indicated by shifts in centroid. However, the centroid is easy to calculate and to compare between different time bins, and more importantly, it can reflect asymmetrical variation in occupation, which is related to selective extinction or origination[31]. A major advantage of reporting centroid-related metrics is that centroids can be calculated using all PCO axes, not just the first two or three[74]. By contrast, a direct observation of morphospace (usually constructed from the first two or three axes) cannot show information on other axes, and this cannot be ignored when the first few axes explain such a low proportion of the variance. Therefore, the centroid-related metric is a valuable complement to the morphospace.

For every order, we first partitioned the morphospace according to the presence or absence table of taxa and calculated the position of the centroid of all time bins using all PCO axes. Next, Euclidean distances between centroids in different time bins were calculated. This calculation generated a distance matrix of all time bins; the distances between centroids of two adjacent bins were extracted as the 'shifts in centroid'. Finally, we used the PCOA (equivalent to principal component analysis because the Euclidean distance was used) to visualize this distance matrix (that is, the 'centroid space'). No further calculation was performed on the centroid space.

It should be noted that the centroid space does not directly show the morphological forms in a time bin. However, it clearly illustrates the relative similarities (or distances) of morphological composition (or 'mean' morphology) among time bins, in other words, in which bins the morphological compositions are closer to each other.

To examine whether the morphospace occupation in adjacent time bins is significantly different, we also performed the permutational multivariate ANOVA test. The analysed data are coordinates of taxa in the morphospace, and again, all PCO axes were considered.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Occurrence data were downloaded from the Paleobiology Database (https://paleobiodb.org/). The added Permian–Jurassic fossil occurrences were from ref. 10. All occurrence data and character matrices analysed are available via Zenodo at https://doi.org/10.5281/zenodo.11559129 (ref. 95).

Code availability

All R scripts used to conduct the analyses are available via Zenodo at https://doi.org/10.5281/zenodo.11559129 (ref. 95).

References

1. Mannion, P. D. et al. Climate constrains the evolutionary history and biodiversity of crocodylians. *Nat. Commun.* **6**, 8438 (2015).
2. Stubbs, T. L. et al. Ecological opportunity and the rise and fall of crocodylomorph evolutionary innovation. *Proc. R. Soc. B* **288**, 20210069 (2021).
3. Payne, A. R., Mannion, P. D., Lloyd, G. T. & Davis, K. E. Decoupling speciation and extinction reveals both abiotic and biotic drivers shaped 250 million years of diversity in crocodile-line archosaurs. *Nat. Ecol. Evol.* **8**, 121–132 (2023).
4. Steele-Petrović, H. M. The physiological differences between articulate brachiopods and filter-feeding bivalves as a factor in the evolution of marine level-bottom communities. *Palaeontology* **22**, 101–134 (1979).
5. Gould, S. J. & Calloway, C. B. Clams and brachiopods—ships that pass in the night. *Paleobiology* **6**, 383–396 (1980).
6. Thayer, C. W. Brachiopods versus mussels: competition, predation, and palatability. *Science* **228**, 1527–1528 (1985).
7. Sepkoski, J. J. Jr. in *Evolutionary Paleobiology* (eds Jablonski, D. et al.) 211–255 (Univ. Chicago Press, 1996).
8. Payne, J. L., Heim, N. A., Knope, M. L. & McClain, C. R. Metabolic dominance of bivalves predates brachiopod diversity decline by more than 150 million years. *Proc. R. Soc. B* **281**, 20133122 (2014).
9. Liow, L. H., Reitan, T. & Harnik, P. G. Ecological interactions on macroevolutionary time scales: clams and brachiopods are more than ships that pass in the night. *Ecol. Lett.* **18**, 1030–1039 (2015).
10. Guo, Z., Flannery-Sutherland, J. T., Benton, M. J. & Chen, Z. Q. Bayesian analyses indicate bivalves did not drive the downfall of brachiopods following the Permian–Triassic mass extinction. *Nat. Commun.* **14**, 5566 (2023).
11. Fraiser, M. L. & Bottjer, D. J. When bivalves took over the world. *Paleobiology* **33**, 397–413 (2007).
12. Raup, D. M. & Sepkoski, J. J. Jr Mass extinctions in the marine fossil record. *Science* **215**, 1501–1503 (1982).
13. Marshall, C. R. Forty years later: the status of the 'Big Five' mass extinctions. *Camb. Prism Extinction* **1**, e5 (2023).
14. Carlson, S. J. The evolution of Brachiopoda. *Annu. Rev. Earth Planet. Sci.* **44**, 409–438 (2016).
15. Carlson, S. J. A phylogenetic perspective on articulate brachiopod diversity and the Permo-Triassic extinction. In *The Unity of Evolutionary Biology, Proc. 4th International Congress of Systematic and Evolutionary Biology* (ed. Dudley, E. C.) 119–142 (Dioscorides Press, 1991).
16. Sepkoski, J. J. Jr. A compendium of fossil marine animal genera. *Bull. Am. Paleontol.* **363**, 1–560 (2002).
17. Chen, Z. Q., Kaiho, K. & George, A. D. Survival strategy of brachiopod fauna from the end-Permian mass extinction. *Palaeogeogr. Palaeoclimatol. Palaeoecol.* **224**, 232–269 (2005).
18. Chen, Z. Q., Kaiho, K. & George, A. D. Early Triassic recovery of the brachiopod faunas from the end-Permian mass extinction: a global review. *Palaeogeogr. Palaeoclimatol. Palaeoecol.* **224**, 270–290 (2005).
19. Clapham, M. E. et al. Assessing the ecological dominance of Phanerozoic marine invertebrates. *Palaios* **21**, 431–441 (2006).
20. Stanley, S. M. What has happened to the articulate brachiopods? *Geol. Soc. Am. Abstr. Prog.* **6**, 966–967 (1974).
21. Thayer, C. W. Biological bulldozers and the evolution of marine benthic communities. *Science* **203**, 458–461 (1979).
22. Valentine, J. W. & Jablonski, D. Larval adapations and patterns of brachiopod diversity in space and time. *Evolution* **37**, 1052–1061 (1983).
23. Zezina, O. N. Biogeography of the recent brachiopods. *Paleontol. J.* **42**, 830–858 (2008).
24. Erwin, D. H. Disparity: morphological pattern and developmental context. *Palaeontology* **50**, 57–73 (2007).
25. Foote, M. Morphological diversity in the evolutionary radiation of Paleozoic and post-Paleozoic crinoids. *Paleobiology* **25**, 1–116 (1999).
26. Foote, M. Ecological controls on the evolutionary recovery of post-Paleozoic crinoids. *Science* **274**, 1492–1495 (1996).
27. Foote, M. Morphological disparity in Ordovician–Devonian crinoids and the early saturation of morphological space. *Paleobiology* **20**, 320–344 (1994).
28. Foote, M. Discordance and concordance between morphological and taxonomic diversity. *Paleobiology* **19**, 185–204 (1993).

29. Villier, L. & Korn, D. Morphological disparity of ammonoids and the mark of Permian mass extinctions. *Science* **306**, 264–266 (2004).
30. Bapst, D. W., Bullock, P. C., Melchin, M. J., Sheets, H. D. & Mitchell, C. E. Graptoloid diversity and disparity became decoupled during the Ordovician mass extinction. *Proc. Natl Acad. Sci. USA* **109**, 3428–3433 (2012).
31. Korn, D., Hopkins, M. J. & Walton, S. A. Extinction space—a method for the quantification and classification of changes in morphospace across extinction boundaries. *Evolution* **67**, 2795–2810 (2013).
32. Benton, M. J., Forth, J. & Langer, M. C. Models for the rise of the dinosaurs. *Curr. Biol.* **24**, R87–R95 (2014).
33. Anderson, P. S., Friedman, M., Brazeau, M. D. & Rayfield, E. J. Initial radiation of jaws demonstrated stability despite faunal and environmental change. *Nature* **476**, 206–209 (2011).
34. Dumont, E. R. et al. Morphological innovation, diversification and invasion of a new adaptive zone. *Proc. R. Soc. B* **279**, 1797–1805 (2012).
35. Stubbs, T. L. & Benton, M. J. Ecomorphological diversifications of Mesozoic marine reptiles: the roles of ecological opportunity and extinction. *Paleobiology* **42**, 547–573 (2016).
36. Novack-Gottshall, P. M. et al. Morphological volatility precedes ecological innovation in early echinoderms. *Nat. Ecol. Evol.* **6**, 263–272 (2022).
37. Hughes, M., Gerber, S. & Wills, M. A. Clades reach highest morphological disparity early in their evolution. *Proc. Natl Acad. Sci. USA* **110**, 13875–13879 (2013).
38. Ciampaglio, C. N. Measuring changes in articulate brachiopod morphology before and after the Permian mass extinction event: do developmental constraints limit morphological innovation? *Evol. Dev.* **6**, 260–274 (2004).
39. Harper, D. A. T. & Gallagher, E. Diversity, disparity and distributional patterns amongst the orthide brachiopod groups. *J. Czech Geol. Soc.* **46**, 87–93 (2001).
40. Sclafani, J. A., Congreve, C. R., Krug, A. Z. & Patzkowsky, M. E. Effects of mass extinction and recovery dynamics on long-term evolutionary trends: a morphological study of Strophomenida (Brachiopoda) across the Late Ordovician mass extinction. *Paleobiology* **44**, 603–619 (2018).
41. Guo, Z., Chen, Z. Q. & Harper, D. A. T. Phylogenetic and ecomorphologic diversifications of spiriferinid brachiopods after the end-Permian extinction. *Paleobiology* **46**, 495–510 (2020).
42. Smith, L. H. & Bunje, P. M. Morphologic diversity of inarticulate brachiopods through the Phanerozoic. *Paleobiology* **25**, 396–408 (1999).
43. Liang, Y. et al. Evolutionary contingency in lingulid brachiopods across mass extinctions. *Curr. Biol.* **33**, 1565–1572 (2023).
44. Lee, D. E. et al. in *Treatise on Invertebrate Paleontology. Part H, Brachiopoda, Revised* Vol. 5 (ed. Kaesler, R. L.) 1965–2250 (Geological Society of America and Univ. Kansas, 2006).
45. Vörös, A., Kocsis, Á. T. & Pálfy, J. Demise of the last two spire-bearing brachiopod orders (Spiriferinida and Athyridida) at the Toarcian (Early Jurassic) extinction event. *Palaeogeogr. Palaeoclimatol. Palaeoecol.* **457**, 233–241 (2016).
46. Baeza-Carratalá, J. F., García Joral, F., Giannetti, A. & Tent-Manclús, J. E. Evolution of the last koninckinids (Athyridida, Koninckinidae), a precursor signal of the Early Toarcian mass extinction event in the Western Tethys. *Palaeogeogr. Palaeoclimatol. Palaeoecol.* **429**, 41–56 (2015).
47. Jablonski, D. Survival without recovery after mass extinctions. *Proc. Natl Acad. Sci. USA* **99**, 8139–8144 (2002).
48. Barnes, B. D., Sclafani, J. A. & Zaffos, A. Dead clades walking are a pervasive macroevolutionary pattern. *Proc. Natl Acad. Sci. USA* **118**, e2019208118 (2021).
49. Piazza, V., Ullmann, C. V. & Aberhan, M. Temperature-related body size change of marine benthic macroinvertebrates across the Early Toarcian Anoxic Event. *Sci. Rep.* **10**, 4675 (2020).
50. Baeza-Carratalá, J. F. & García Joral, F. Linking Western Tethyan Rhynchonellide morphogroups to the key post-Palaeozoic extinction and turnover events. *Palaeogeogr. Palaeoclimatol. Palaeoecol.* **553**, 109791 (2020).
51. Kaesler, R. L. & Selden, P. A. *Treatise on Invertebrate Paleontology, Part H (Revised), Brachiopoda* Vols. 1–6 (Geological Society of America and Univ. Kansas, 1997–2007).
52. Stanley, S. M. Evidence that more than a third of Paleozoic articulate brachiopod genera (Strophomenata) lived infaunally. *Paleobiology* **46**, 405–433 (2020).
53. Ager, D. V. The adaptation of Mesozoic brachiopods to different environments. *Palaeogeogr. Palaeoclimatol. Palaeoecol.* **1**, 143–172 (1965).
54. Cooper, G. A. Homeomorphy in recent deep-sea brachiopods. *Smithson. Contrib. Paleobiol.* **11**, 1–25 (1972).
55. Vörös, A. The smooth brachiopods of the Mediterranean Jurassic: refugees or invaders? *Palaeogeogr. Palaeoclimatol. Palaeoecol.* **223**, 222–242 (2005).
56. Logan, A. in *Treatise on Invertebrate Paleontology. Part H, Brachiopoda, Revised* Vol. 6 (ed. Selden, P. A.) 3082–3115 (Geological Society of America and Univ. Kansas, 2007).
57. Peck, L. S. Body volumes and internal space constraints in articulate brachiopods. *Lethaia* **25**, 383–390 (1992).
58. McGowan, A. J. Ammonoid taxonomic and morphologic recovery patterns after the Permian–Triassic. *Geology* **32**, 665–668 (2004).
59. Chen, Z. Q. & Benton, M. J. The timing and pattern of biotic recovery following the end-Permian mass extinction. *Nat. Geosci.* **5**, 375–383 (2012).
60. Song, H., Wignall, P. B. & Dunhill, A. M. Decoupled taxonomic and ecological recoveries from the Permo-Triassic extinction. *Sci. Adv.* **4**, eaat5091 (2018).
61. Jaecks, G. S. & Carlson, S. J. How phylogenetic inference can shape our view of heterochrony: examples from thecideide brachiopods. *Paleobiology* **27**, 205–225 (2001).
62. Baker, P. G. in *Treatise on Invertebrate Paleontology. Part H, Brachiopoda, Revised* Vol. 5 (ed. Kaesler, R. L.) 1938–1964 (Geological Society of America and Univ. Kansas, 2006).
63. Alvarez, F. & Rong, J. Y. in *Treatise on Invertebrate Paleontology. Part H, Brachiopoda, Revised* Vol. 5 (ed. Kaesler, R. L.) 1475–1616 (Geological Society of America and Univ. Kansas, 2002).
64. Sepkoski, J. J. Jr A factor analytic description of the Phanerozoic marine fossil record. *Paleobiology* **7**, 36–53 (1981).
65. Stanley, S. M. An analysis of the history of marine animal diversity. *Paleobiology* **33**, 1–55 (2007).
66. Mondal, S. & Harries, P. J. The effect of taxonomic corrections on Phanerozoic generic richness trends in marine bivalves with a discussion on the clade's overall history. *Paleobiology* **42**, 157–171 (2016).
67. Losos, J. B. Adaptive radiation, ecological opportunity, and evolutionary determinism. *Am. Nat.* **175**, 623–639 (2010).
68. Ager, D. V. Why the rhynchonellid brachiopods survived and the spiriferids did not: a suggestion. *Palaeontology* **30**, 853–857 (1987).
69. Jablonski, D. Approaches to macroevolution: 1. General concepts and origin of variation. *Evol. Biol.* **44**, 427–450 (2017).
70. Erwin, D. H. Novelty and innovation in the history of life. *Curr. Biol.* **25**, R930–R940 (2015).
71. Peng, Y. & Shi, G. R. New Early Triassic Lingulidae (Brachiopoda) genera and species from South China. *Alcheringa* **32**, 149–170 (2008).
72. Posenato, R. Systematics of lingulide brachiopods from the end-Permian mass extinction interval. *Riv. Ital. Paleontol. Stratigr.* **122**, 85–108 (2016).

73. Bitner, M. A. & Emig, C. C. On the type species of *Lingularia*, and description of *Eolingularia* n. gen. *Carnets Geol.* **16**, 543–555 (2016).
74. Cole, S. R. & Hopkins, M. J. Selectivity and the effect of mass extinctions on disparity and functional ecology. *Sci. Adv.* **7**, eabf4072 (2021).
75. Gradstein, F. M., Ogg, J. G., Schmitz, M. D. & Ogg, G. M. *Geologic Time Scale 2020* (Elsevier, 2020).
76. Flannery-Sutherland, J. T., Raja, N. B., Kocsis, Á. T. & Kiessling, W. fossilbrush: an R package for automated detection and resolution of anomalies in palaeontological occurrence data. *Methods Ecol. Evol.* **13**, 2404–2418 (2022).
77. Raup, D. M. Biases in the fossil record of species and genera. *Bull. Carnegie Mus. Nat. Hist.* **13**, 85–91 (1979).
78. Schreiber, H. A., Bitner, M. A. & Carlson, S. J. Morphological analysis of phylogenetic relationships among extant rhynchonellide brachiopods. *J. Paleontol.* **87**, 550–569 (2013).
79. Guo, Z., Chen, Z. Q., Harper, D. A. T. & Huang, Y. G. Permian–Triassic phylogenetic and morphologic evolution of rhynchonellide brachiopods. *Paleobiology* **48**, 99–119 (2022).
80. Carlson, S. J. & Fitzgerald, P. C. Sampling taxa, estimating phylogeny and inferring macroevolution: an example from Devonian terebratulide brachiopods. *Earth Environ. Sci. Trans. R. Soc. Edinb.* **98**, 311–325 (2008).
81. Matzke, N. J. & Irmis, R. B. Including autapomorphies is important for paleontological tip-dating with clocklike data, but not with non-clock data. *PeerJ* **6**, e4553 (2018).
82. Hopkins, M. J. & St John, K. A new family of dissimilarity metrics for discrete character matrices that include inapplicable characters and its importance for disparity studies. *Proc. R. Soc. B* **285**, 20181784 (2018).
83. Lloyd, G. T. Estimating morphological diversity and tempo with discrete character-taxon matrices: implementation, challenges, progress, and future directions. *Biol. J. Linn. Soc.* **118**, 131–151 (2016).
84. Flannery Sutherland, J. T., Moon, B. C., Stubbs, T. L. & Benton, M. J. Does exceptional preservation distort our view of disparity in the fossil record? *Proc. R. Soc. B* **286**, 20190091 (2019).
85. Gerber, S. Use and misuse of discrete character data for morphospace and disparity analyses. *Palaeontology* **62**, 305–319 (2019).
86. Paradis, E. & Schliep, K. ape 5.0: an environment for modern phylogenetics and evolutionary analyses in R. *Bioinformatics* **35**, 526–528 (2019).
87. Cailliez, F. The analytical solution of the additive constant problem. *Psychometrika* **48**, 305–308 (1983).
88. Nordén, K. K., Stubbs, T. L., Prieto-Márquez, A. & Benton, M. J. Multifaceted disparity approach reveals dinosaur herbivory flourished before the end-Cretaceous mass extinction. *Paleobiology* **44**, 620–637 (2018).
89. Ciampaglio, C. N., Kemp, M. & McShea, D. W. Detecting changes in morphospace occupation patterns in the fossil record: characterization and analysis of measures of disparity. *Paleobiology* **27**, 695–715 (2001).
90. Guillerme, T., Puttick, M. N., Marcy, A. E. & Weisbecker, V. Shifting spaces: which disparity or dissimilarity measurement best summarize occupancy in multidimensional spaces? *Ecol. Evol.* **10**, 7261–7275 (2020).
91. Foote, M. Rarefaction analysis of morphological and taxonomic diversity. *Paleobiology* **18**, 1–16 (1992).
92. Kowalewski, M. & Novack-Gottshall, P. M. Resampling methods in paleontology. *Paleontolog. Soc. Pap.* **16**, 19–54 (2010).
93. Guillerme, T. dispRity: a modular R package for measuring disparity. *Methods Ecol. Evol.* **9**, 1755–1763 (2018).
94. Oksanen, J. et al. vegan: community ecology package. R package version 2.6-4 https://CRAN.R-project.org/package=vegan (2022).
95. Guo, Z., Benton, M. J., Thomas, T. L. & Chen, Z. Q. Supplementary data Guo et al. Brachiopod morphological evolution. *Zenodo* https://doi.org/10.5281/zenodo.11559129 (2024).

Acknowledgements

We are grateful to contributors of the PBDB. Z. Y. Li, X. Luo, V. J. Radulović, B. V. Radulović, A. E. Schrøder, S. Calzada Badia, D. A. B. MacFarlan, A. Vörös, X. M. Lu, Y.P. Sun, S. Lee, T. A. Grunt, J. Robinson and D. E. Lee are acknowledged for providing important literature. D. A. B. MacFarlan is thanked for the discussion of Zealandia brachiopods. This study was funded by a grant (42488201) of basic science centre programme and two grants (41930322, 92055212) of NSFC (Z.-Q.C.), as well as grant NE/X013111/1 from NERC (M.J.B.).

Author contributions

Z.-Q.C. and Z.G. designed the research. Z.G. collected morphological data. Z.G. and T.L.S. performed the analysis. All authors discussed the results. Z.G. drafted the paper with substantial input from all authors.

Competing interests

The authors declare no competing interests.

Additional information

Extended data is available for this paper at https://doi.org/10.1038/s41559-024-02491-9.

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41559-024-02491-9.

Correspondence and requests for materials should be addressed to Zhong-Qiang Chen.

Peer review information *Nature Ecology & Evolution* thanks Philip Novack-Gottshall, Pedro Monarrez and Judith Sclafani for their contribution to the peer review of this work. Peer reviewer reports are available.

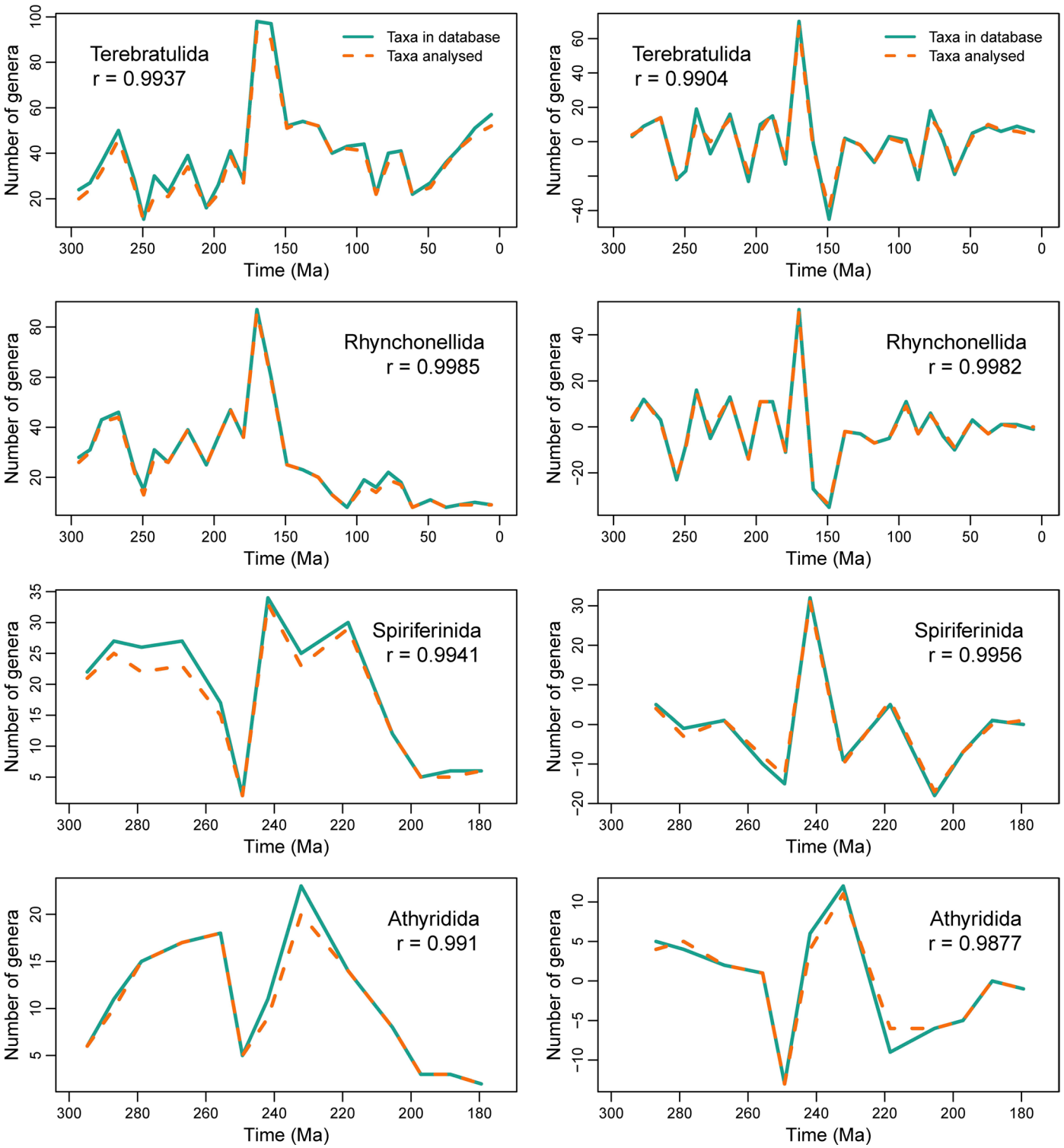
Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

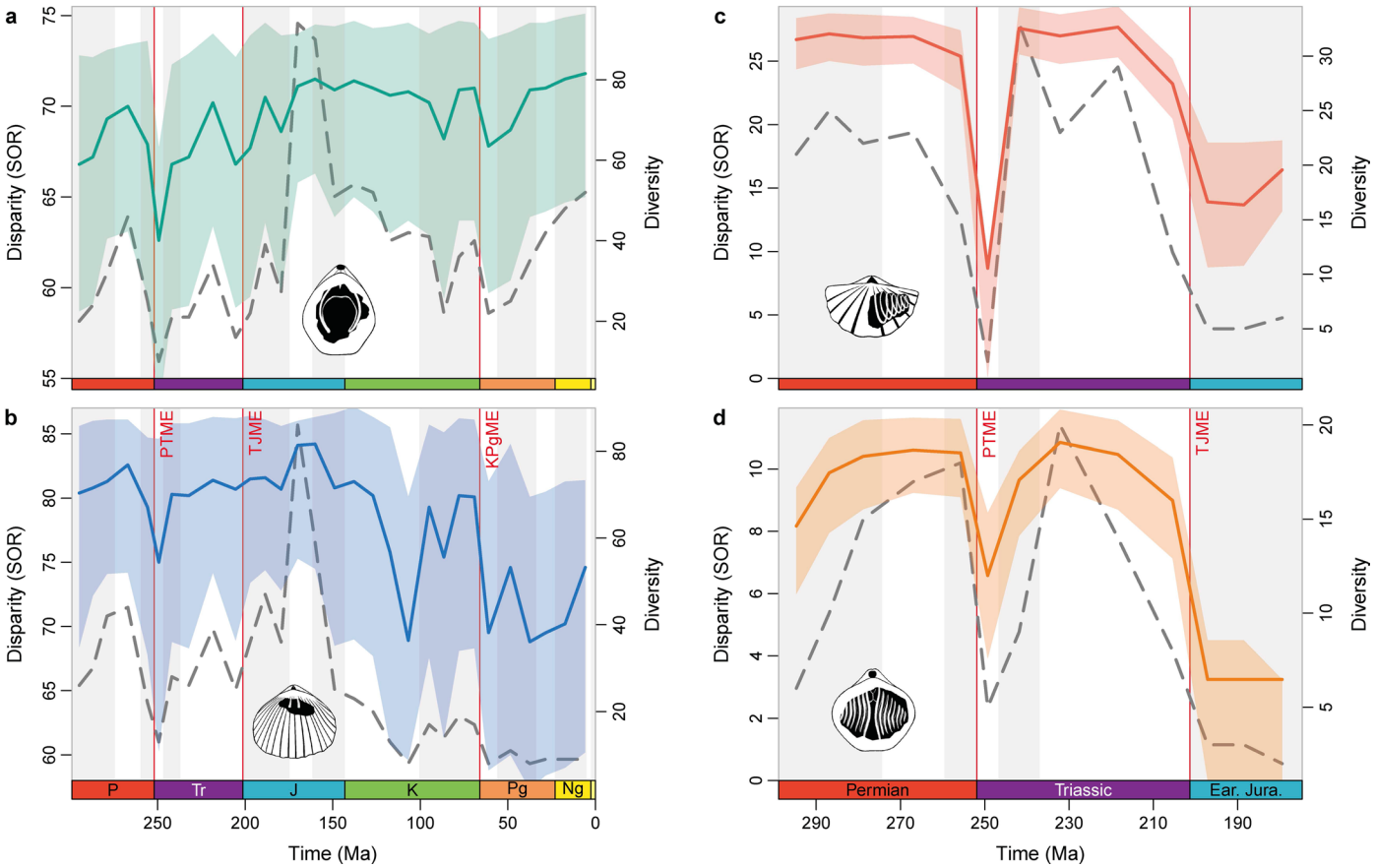
© The Author(s), under exclusive licence to Springer Nature Limited 2024

https://doi.org/10.1038/s41559-024-02491-9



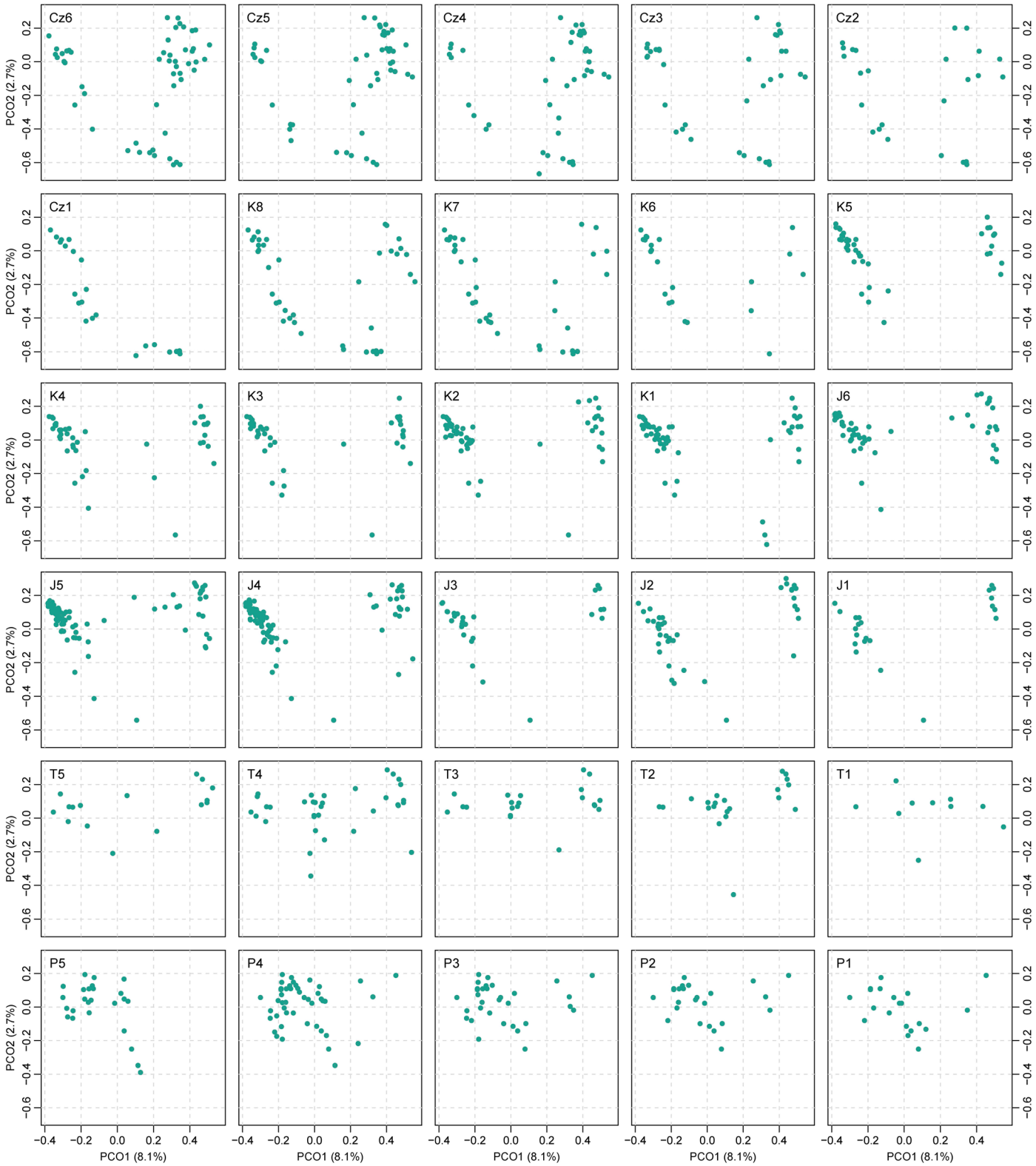
Extended Data Fig. 1 | Diversity of taxa in the revised occurrence database and diversity of taxa included in the morphological analysis. The left panel shows the comparison of raw diversity (sample size $n = 30$ for Rhynchonellida and Terebratulida; $n = 13$ for Spiriferinida and Athyridida). The right panel shows the comparison of detrended (that is, first difference) data. Correlation parameters (Pearson's r) are indicated. All correlations are significant ($P < 0.05$; two-sided).

https://doi.org/10.1038/s41559-024-02491-9



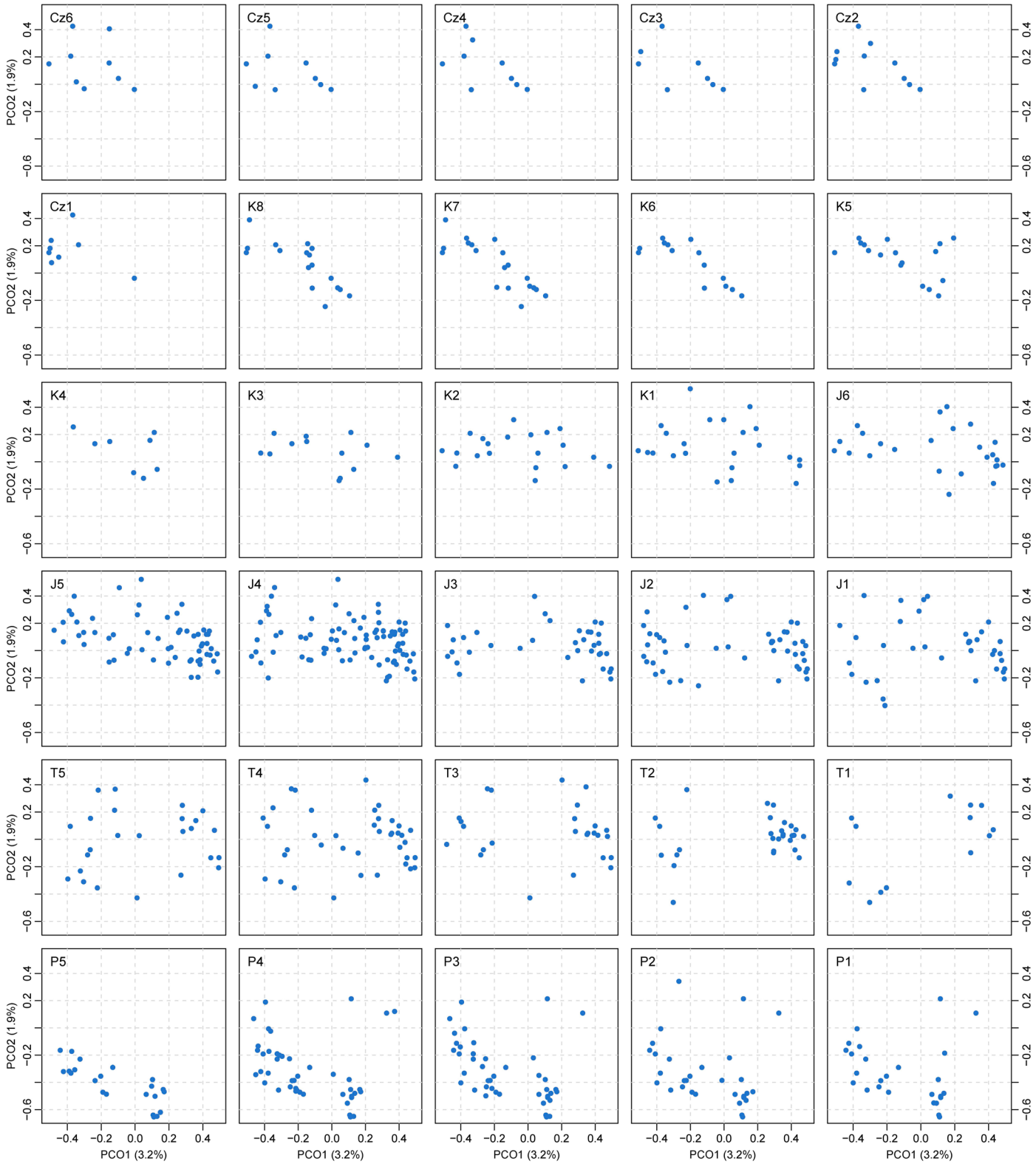
Extended Data Fig. 2 | Disparity and diversity of four orders through time. Disparity is measured by sum of ranges (SOR). The solid lines represent median value of the bootstrapped disparity. The envelopes indicate 95% confidence intervals (0.025 and 0.975 quantiles of bootstrapped values). Diversity (number of genera) is represented by dashed lines. **a**, Terebratulida; **b**, Rhynchonellida; **c**, Spiriferinida; **d**, Athyridida. Abbreviations as in Fig. 2.

https://doi.org/10.1038/s41559-024-02491-9



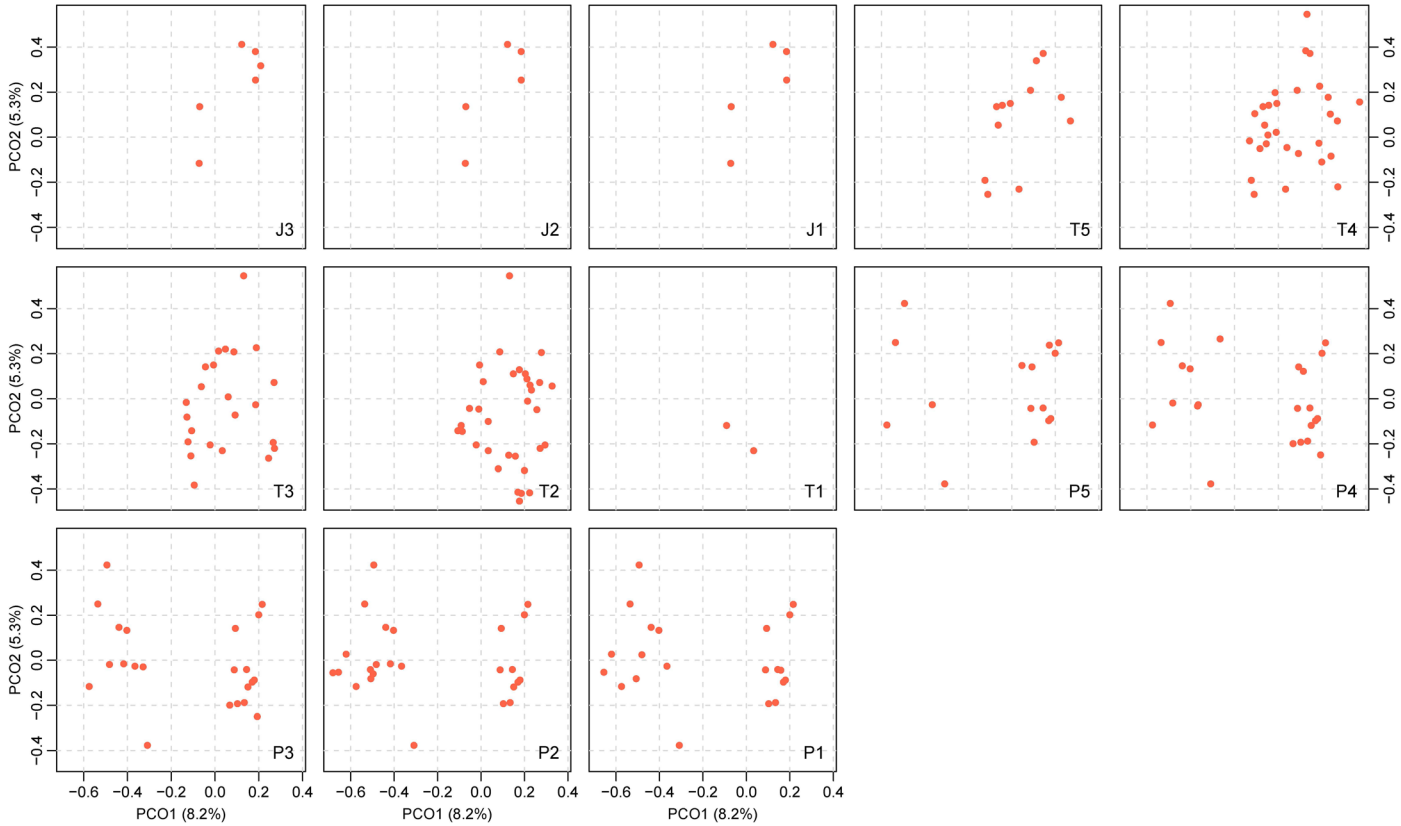
Extended Data Fig. 3 | Morphospace occupation of the order Terebratulida through time. Morphospaces are constructed by the first two axes (PCO1 and PCO2). The number in labels indicates the percentage of variances explained by the axis.

https://doi.org/10.1038/s41559-024-02491-9



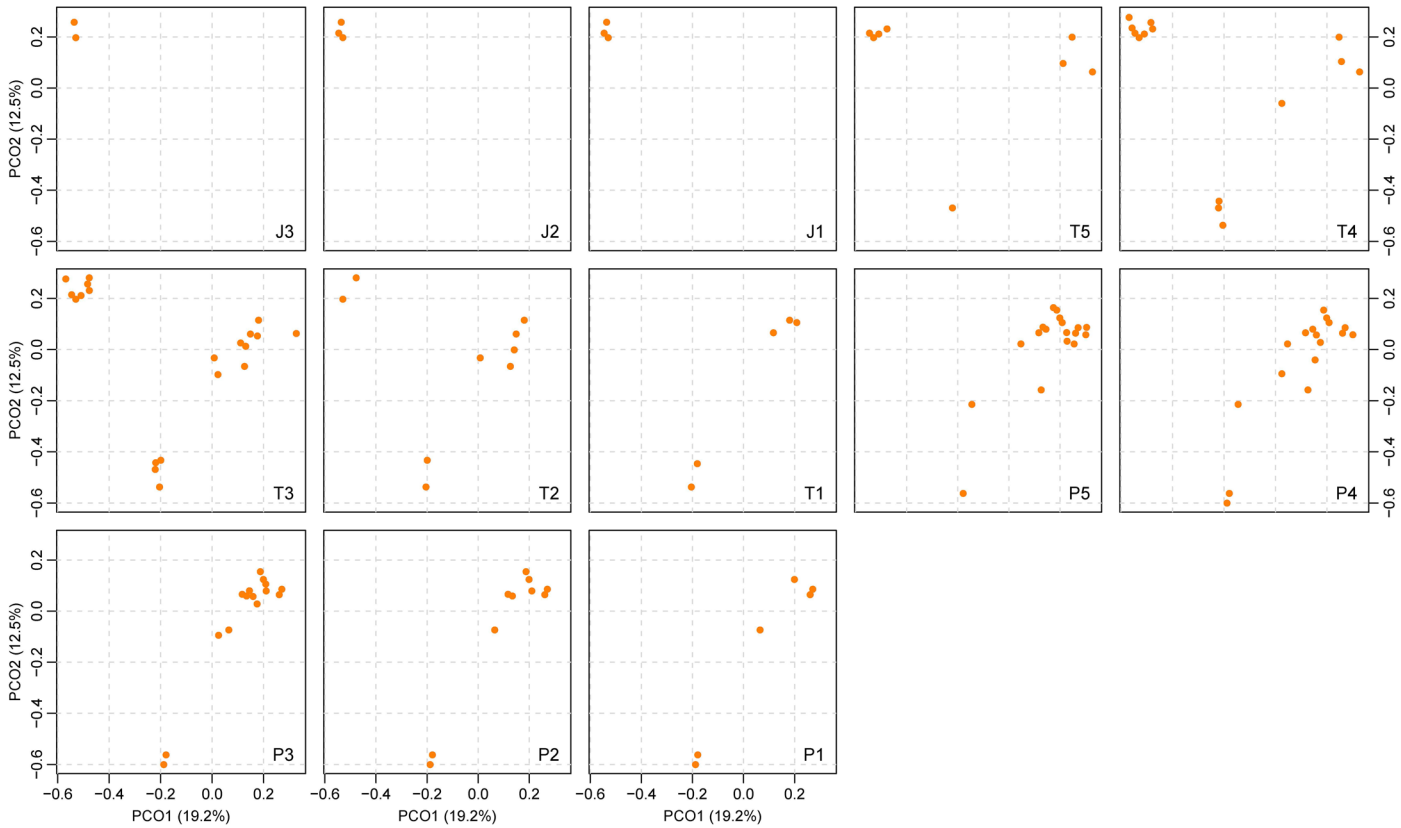
Extended Data Fig. 4 | Morphospace occupation of the order Rhynchonellida through time. Morphospaces are constructed by the first two axes (PCO1 and PCO2). The number in labels indicates the percentage of variances explained by the axis.

https://doi.org/10.1038/s41559-024-02491-9



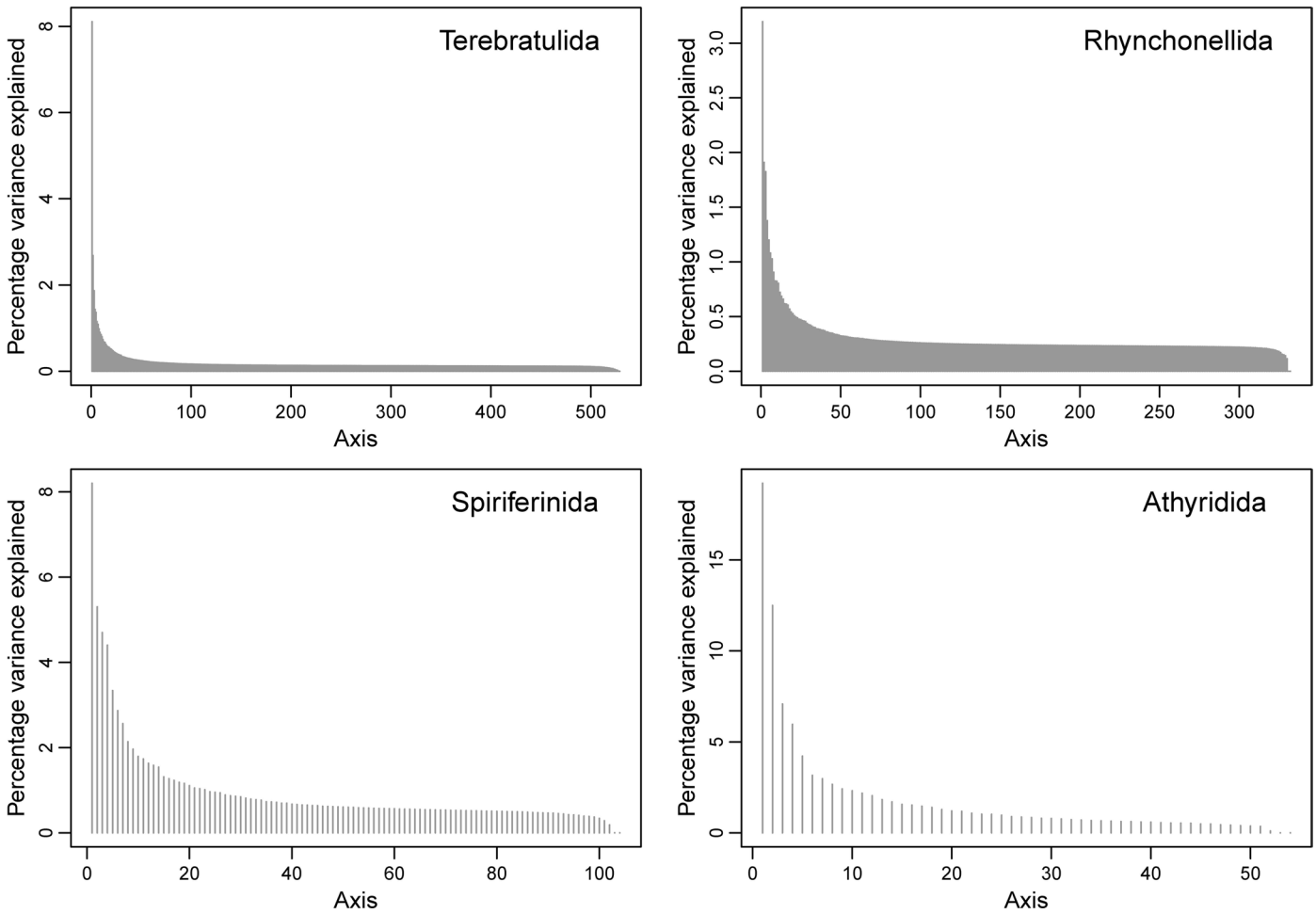
Extended Data Fig. 5 | Morphospace occupation of the order Spiriferinida through time. Morphospaces are constructed by the first two axes (PCO1 and PCO2). The number in labels indicates the percentage of variances explained by the axis.

https://doi.org/10.1038/s41559-024-02491-9



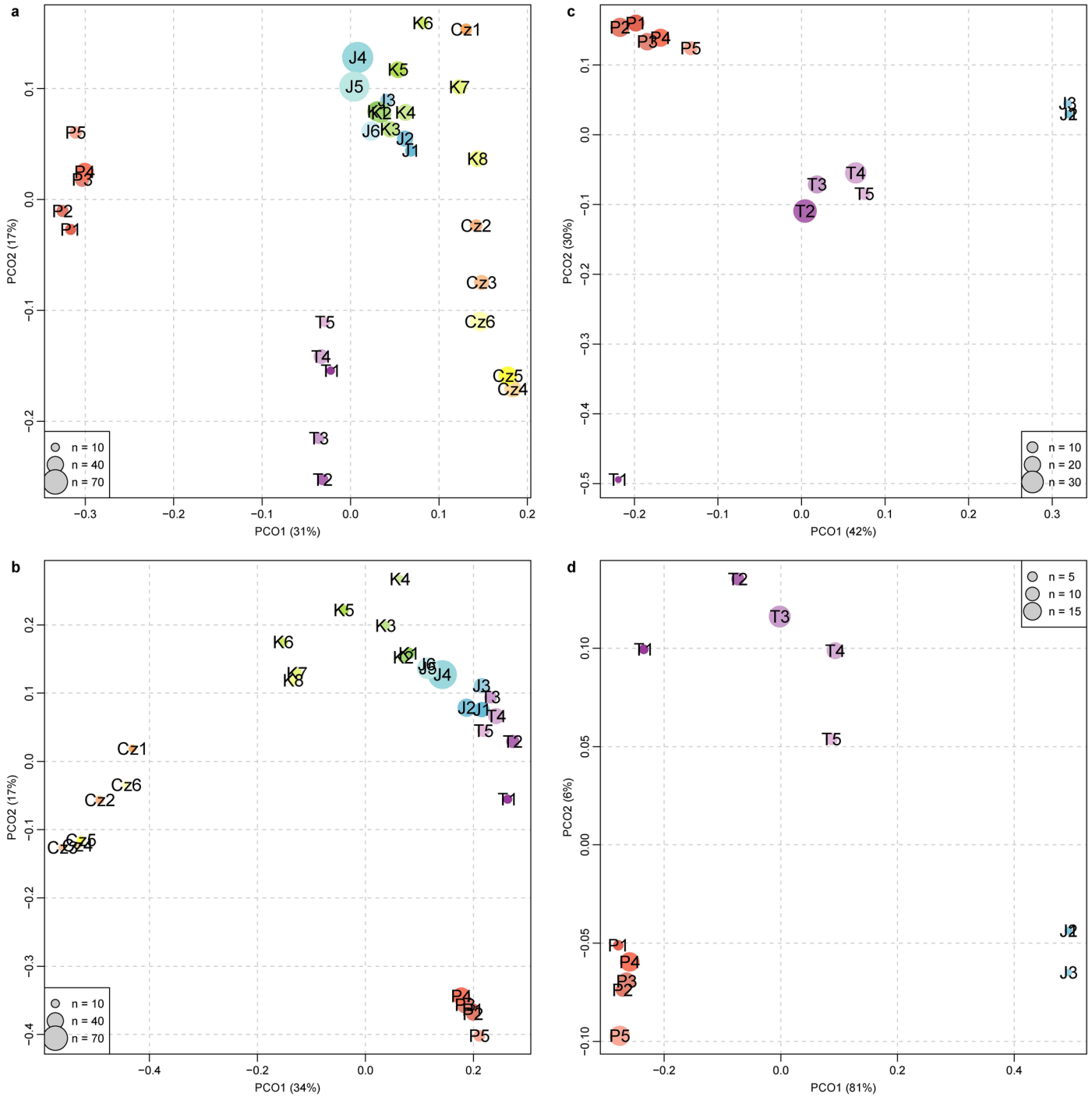
Extended Data Fig. 6 | Morphospace occupation of the order Athyridida through time. Morphospaces are constructed by the first two axes (PCO1 and PCO2). The number in labels indicates the percentage of variances explained by the axis.

https://doi.org/10.1038/s41559-024-02491-9



Extended Data Fig. 7 | Percentage variance explained by each axis of the PCOA of morphological data.

https://doi.org/10.1038/s41559-024-02491-9



Extended Data Fig. 8 | Morphospace centroid spaces of the four orders with bin names labelled. a, Terebratulida; **b**, Rhynchonellida; **c**, Spiriferinida; **d**, Athyridida.

nature portfolio

Corresponding author(s): Zhong-Qiang Chen

Last updated by author(s): Jun 11, 2022

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our Editorial Policies and the Editorial Policy Checklist.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☒	☐ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☒	☐ A description of all covariates tested
☐	☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒ For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐ Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	No software was used for data collection.
Data analysis	All analyses were performed using R (v4.1). Packages employed to perform the analyses are fossilbrush (v1.0.3), ape (v5.6-2), dispRity (v1.7.0), and vegan (v2.6-4). All code used to analyse the data are available in Zenodo (https://doi.org/10.5281/zenodo.11559129).

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

Most data were downloaded from Paleobiology Database (https://paleobiodb.org/). In addition, we added some fossil occurrences collected from original references. All added, raw and revised data analysed in this study are available in Zenodo (https://doi.org/10.5281/zenodo.11559129).

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	This is not a human research.
Reporting on race, ethnicity, or other socially relevant groupings	This is not a human research.
Population characteristics	This is not a human research.
Recruitment	This is not a human research.
Ethics oversight	This is not a human research.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☐ Life sciences ☐ Behavioural & social sciences ☒ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Ecological, evolutionary & environmental sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	This study analysed morphological evolution of four major Meso-Cenozoic brachiopod orders (the Terebratulida, Rhynchonellida, Spiriferinida, and Athyridida) over the past ~300 million years (Permian to Holocene). Morphological characters of more than 1000 genera were collected from primary literature. Principal coordinates analysis was used to construct the morphospace. Based on these results, we reported variation of morphological disparity through time, morphospace occupation, morphospace centroid space, and shifts in centroids of the four orders.
Research sample	The analysed data are morphological characters of more than 1000 brachiopod genera. Occurrence data of brachiopods were used to indicate the presence/absence of genera in discrete time bins.
Sampling strategy	Except for genera with poorly known morphological characters, all genera in the occurrence database downloaded from the Paleobiology Database were included in the study.
Data collection	All morphological characters of brachiopods were collected by the first author from primary literature.
Timing and spatial scale	The timing scale of our study is from Permian to Quaternary. The spatial scale is global.
Data exclusions	We wanted to include as many taxa as possible. However, missing data can affect the position of taxa in the morphospace. Therefore, taxa with more than 20–30% (20% for terebratulides and rhynchonellides; 30% for spiriferinides and athyridides) unknown characters (marked by question mark '?') were discarded from the dataset. At last, only less than 10.5% of genera were discarded. Names of discarded taxa are provided in Supplementary Information.
Reproducibility	All data and code needed to recreate the analyses are available in Zenodo (https://doi.org/10.5281/zenodo.11559129).
Randomization	Taxa were divided into temporal groups based on their stratigraphical age.
Blinding	Blinding was not required because we used all data in the analyses. There is no experiments in which information may influence the result.

Did the study involve field work? ☐ Yes ☒ No

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks	This is not a plant research.
Novel plant genotypes	This is not a plant research.
Authentication	This is not a plant research.